

REPORT

BUILDING AN END-TO-END FUEL PRICE ANALYTICS SYSTEM USING SNOWFLAKE & AIRFLOW

DATA 226 - Group 9

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ABSTRACT

This report presents an end-to-end fuel price analytics system built with Apache Airflow, Snowflake, dbt, Snowflake ML, and Preset. The system collects historical gasoline and diesel prices from the U.S. Energy Information Administration API and current energy market indicators from Yahoo Finance through yfinance. Airflow orchestrates ingestion, Snowflake stores raw and analytics-ready data, dbt creates transformation models for moving averages, volatility, crude oil correlation, and regional comparison, and Snowflake ML generates a 12-week regular gasoline forecast. The final Preset dashboard enables stakeholders to analyze national trends, regional price differences, week-over-week volatility, crude oil market influence, and forecast gasoline prices.

Keywords: Fuel Analytics, ETL, ELT, Apache Airflow, Snowflake, dbt, Snowflake ML, Forecasting, yfinance, EIA API, Data Warehouse, BI Dashboard, SCD Type 2

I. INTRODUCTION

Fuel prices affect households, logistics, delivery, retail, aviation, and supply chain operations. When gasoline and diesel prices rise, consumers and businesses experience higher transportation costs. Because fuel prices change over time and differ across U.S. regions, an automated analytics system is useful for tracking price movement, identifying regional gaps, measuring volatility, and understanding market drivers.

This project builds a complete analytics workflow. Public source data is ingested through Airflow, stored in Snowflake, transformed with dbt, forecast using Snowflake ML, and visualized through an interactive Preset dashboard.

II. PROBLEM STATEMENT

Raw fuel price data is not immediately decision-ready. A user may need to know whether prices are rising, which regions are expensive, whether the market is volatile, whether crude oil explains gasoline price movement, and what prices may look like in the near future.

The project answers five main analytical questions:

1. How are U.S. gasoline and diesel prices changing over time?
2. Which U.S. regions are more expensive or cheaper than the national average?
3. How volatile are fuel prices from week to week?
4. How does WTI crude oil relate to retail gasoline prices?
5. What is the 12-week forecast for regular gasoline prices?

III. SYSTEM REQUIREMENTS & SPECIFICATIONS

A. Functional Requirements

The system shall:

1. Retrieve historical weekly national fuel price data from the U.S. Energy Information Administration API.
2. Retrieve regional gasoline price data by PADD region from the EIA API.
3. Extract energy market data, including WTI crude, Brent crude, XLE, and UGA prices using Yahoo Finance.
4. Transform raw API and market responses into structured tabular records.
5. Load national fuel prices into the Snowflake table RAW.FUEL_PRICES.
6. Load regional gasoline prices into RAW.REGIONAL_FUEL_PRICES.
7. Load crude oil and energy market data into RAW.ENERGY_MARKET_PRICES.
8. Use MERGE/upsert logic to prevent duplicate rows during repeated pipeline runs.
9. Create dbt analytics models for moving averages, volatility, crude correlation, and regional comparison.
10. Create a clean ML training view using WEEK_DATE and REGULAR_GASOLINE_PRICE.

11. Train a Snowflake ML Forecast model on historical regular gasoline prices.
12. Generate a 12-week gasoline price forecast with prediction intervals.
13. Store forecast results in ADHOC.FUEL_PRICE_FORECAST.
14. Produce a unified analytics table, ANALYTICS.FUEL_PRICE_FINAL, combining historical and forecast data.
15. Build an interactive Preset dashboard showing fuel trends, forecasts, volatility, regional comparison, and crude oil relationships.
16. Execute ETL, dbt, and ML pipelines automatically using scheduled Airflow DAGs.

B. Non-Functional Requirements

1. The system shall complete ETL execution within a reasonable time under normal network and Snowflake warehouse conditions.
2. Airflow tasks shall support retries to handle temporary API, network, or Snowflake connection failures.
3. Credentials shall be stored securely using Airflow Connections and environment variables.
4. Snowflake operations shall use structured schemas to separate raw, adhoc, analytics, and dbt objects.
5. The pipeline shall be modular so that additional fuel types, regions, or market tickers can be added with minimal code changes.
6. The dashboard shall be interactive and allow users to filter by date range, record type, and price direction.
7. The architecture shall be maintainable, scalable, and reproducible using Docker, Airflow, Snowflake, dbt, and Preset.

C. System Limitations

1. Forecast accuracy depends on the quality and completeness of EIA historical fuel price data.
2. Yahoo Finance market data may contain missing values, delayed prices, or ticker availability issues.
3. Snowflake ML Forecast is primarily time-series based and may not fully capture external economic or geopolitical events.
4. API rate limits or network issues may temporarily interrupt data extraction.
5. The current implementation focuses mainly on U.S. national and regional gasoline price analysis.
6. Airflow is deployed locally through Docker rather than a production cloud-managed orchestration service.

IV. DATA SOURCES

The project uses two real-world data sources as required by the project specification: one historical/archive dataset and one current/near-real-time dataset.

A. Dataset 1 - EIA Open Data API (Historical Fuel Prices)

Provider: [U.S. Energy Information Administration API Endpoint](#)

EIA API key (stored in Airflow as Variable: eia_api_key)

Update Frequency: Weekly, typically released each Wednesday

The EIA API provides the official U.S. government record of weekly retail fuel prices. Two categories of data are pulled.

National Price Series (latest 200 weekly records): The following EIA series identifiers are used to extract weekly national average retail prices in USD per gallon:

1. Regular Gasoline: EMM_EPMR_PTE_NUS_DPG
2. Midgrade Gasoline: EMM_EPMM_PTE_NUS_DPG
3. Premium Gasoline: EMM_EPMP_PTE_NUS_DPG
4. Diesel: EMD_EPD2D_PTE_NUS_DPG

Regional Price Series (latest 104 weekly records per region): Weekly regular gasoline prices by PADD (Petroleum Administration for Defense Districts) region using the following series identifiers:

1. East Coast: EMM_EPMR_PTE_R10_DPG
2. Midwest: EMM_EPMR_PTE_R20_DPG
3. Gulf Coast: EMM_EPMR_PTE_R30_DPG
4. Rocky Mountain: EMM_EPMR_PTE_R40_DPG
5. West Coast: EMM_EPMR_PTE_R50_DPG
6. California: EMM_EPMR_PTE_SCA_DPG

Data Type	Full Name	Records	Snowflake Target
National Prices	4 fuel types	200	RAW.FUEL_PRICES
Regional prices	6 PADD regions	104/reg	RAW.REGIONAL_FUEL_PRICES

Table 1: EIA Data Summary

B. Dataset 2 - Yahoo Finance via yfinance (Real-Time Market Data)

Provider: Yahoo Finance, accessed via the yfinance Python library

Update Frequency: Daily after market close

History Pulled: 4 years of weekly OHLCV data (Open, High, Low, Close, Volume)

This dataset provides current and historical energy market indicators. It is classified as a near-real-time data source for this project. It does not represent station-level pump prices; it represents financial market instruments tied to crude oil, gasoline futures, and the energy sector. Four tickers are pulled.

Ticker	Full Name	Role in Analytics
CL = F	WTI crude oil futures	Primary US crude benchmark, Key cost driver for gasoline production
BZ = F	Brent Crude Oil Futures	Global crude benchmark reflects international market pressure
XLE	Energy Select Sector ETF	Tracks ExxonMobil, Chevron, and other major energy companies
UGA	United States Gasoline Fund ETF	Tracks gasoline futures, Direct market proxy for pump price direction

Table 2: Finance Ticker Summary

Rationale for CL=F in Correlation Analysis: Crude oil (WTI, ticker CL=F) is the primary raw material used to refine gasoline and diesel. When crude oil futures prices rise, refinery input costs increase, which typically results in higher retail pump prices within a lag window of one to several weeks. The dbt model CRUDE_CORRELATION joins EIA pump prices with CL=F closing prices to quantify this relationship.

V. SYSTEM ARCHITECTURE

The system uses two source layers, one warehouse layer, an ELT transformation layer, a forecasting layer, and a BI layer. EIA provides historical pump price data, while Yahoo Finance provides current market context through crude oil and energy tickers.

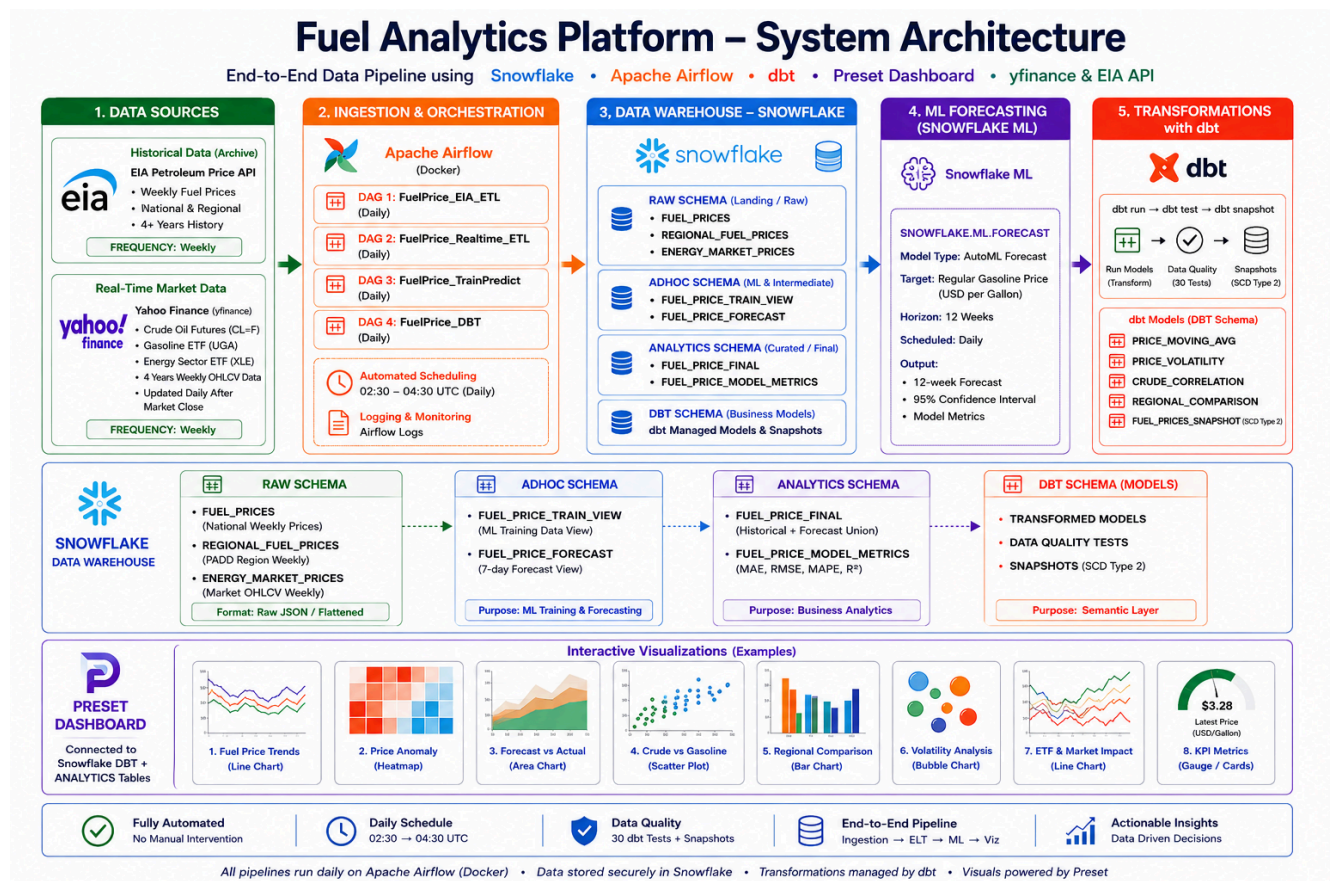


FIGURE 1: Architecture Diagram

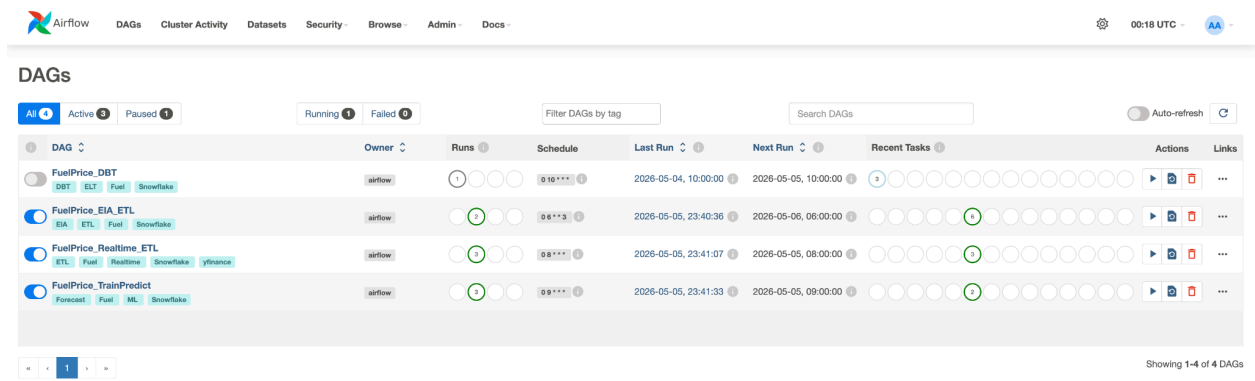


FIGURE 2: The DAG's pipelines in Airflow

DAG Execution Schedule

DAG ID	Schedule	UTC Time	Depends On
FuelPrice_EIA_ETL	0 6 * * 3	Wed 06:00	EIA API available; runs weekly on EIA update day
FuelPrice_Realtime_ETL	0 8 * * *	Daily 08:00	Yahoo Finance market close (previous day)
FuelPrice_TrainPredict	0 9 * * *	Daily 09:00	FuelPrice_EIA_ETL must have loaded RAW.FUEL_PRICES
FuelPrice_DBT	0 10 * * *	Daily 10:00	FuelPrice_EIA_ETL and FuelPrice_Realtime_ETL must have completed

TABLE 3: Airflow DAG schedule showing execution times and dependencies between pipeline stages.

End-to-End Data Flow Summary

DAG	Data Source	Snowflake Table(s) Loaded	Type
FuelPrice_EIA_ETL	EIA API	RAW.FUEL_PRICES	Historical / Archive
FuelPrice_EIA_ETL	EIA API	RAW.REGIONAL_FUEL_PRICES	Historical / Regional
FuelPrice_Realtime_ETL	Yahoo Finance (yfinance)	RAW.ENERGY_MARKET_PRICES	Current / Near-Real-Time
FuelPrice_TrainPredict	RAW.FUEL_PRICES	ANALYTICS.FUEL_PRICE_FINAL	ML Forecast Output
FuelPrice_DBT	RAW Tables	DBT.PRICE_MOVING_AVG	ELT Analytics
FuelPrice_DBT	RAW Tables	DBT.PRICE_VOLATILITY	ELT Analytics
FuelPrice_DBT	RAW Tables	DBT.CRUDE_CORRELATION	ELT Analytics
FuelPrice_DBT	RAW Tables	DBT.REGIONAL_COMPARISON	ELT Analytics
FuelPrice_DBT	RAW.FUEL_PRICES	DBT.FUEL_PRICES_SNAPSHOT	SCD Type 2 Snapshot

TABLE 4: Complete data flow summary showing DAG, source, target table, and data type for each pipeline stage.

Pipeline 1 - FuelPrice_EIA_ETL

1. Extracts national weekly gasoline and diesel price data from the U.S. Energy Information Administration API.
2. Extracts regional gasoline prices for U.S. PADD regions such as California, West Coast, East Coast, Midwest, Gulf Coast, and Rocky Mountain.
3. Transforms API responses into structured records aligned with the Snowflake schemas.
4. Loads national fuel price data into RAW.FUEL_PRICES.
5. Loads regional fuel price data into RAW.REGIONAL_FUEL_PRICES.
6. Uses a MERGE / upsert strategy to prevent duplicate weekly records.
7. Scheduled weekly every Wednesday at 6:00 AM UTC.

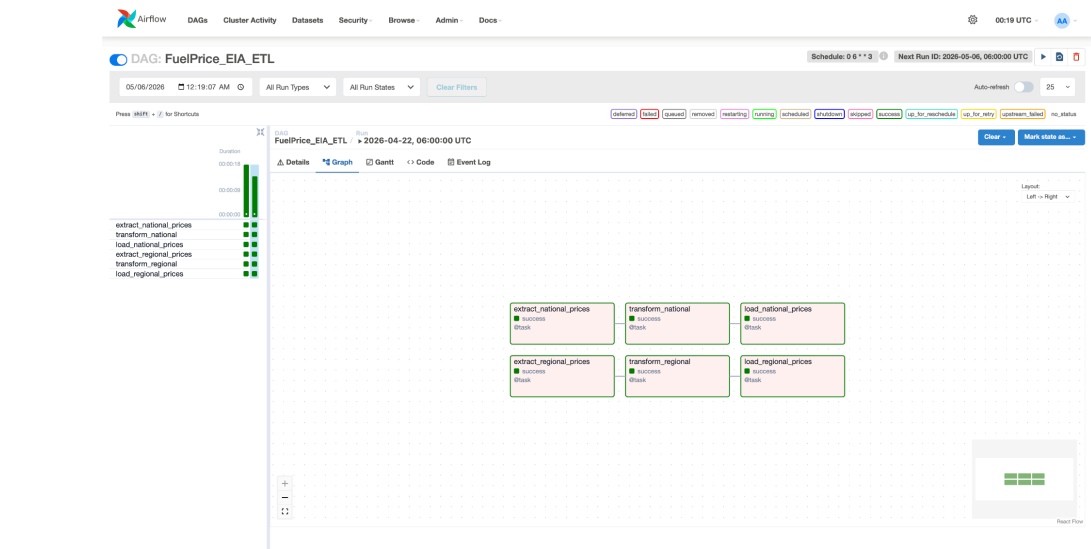


FIGURE 3: Detailed view of Pipeline 1 (FuelPrice_EIA_ETL)

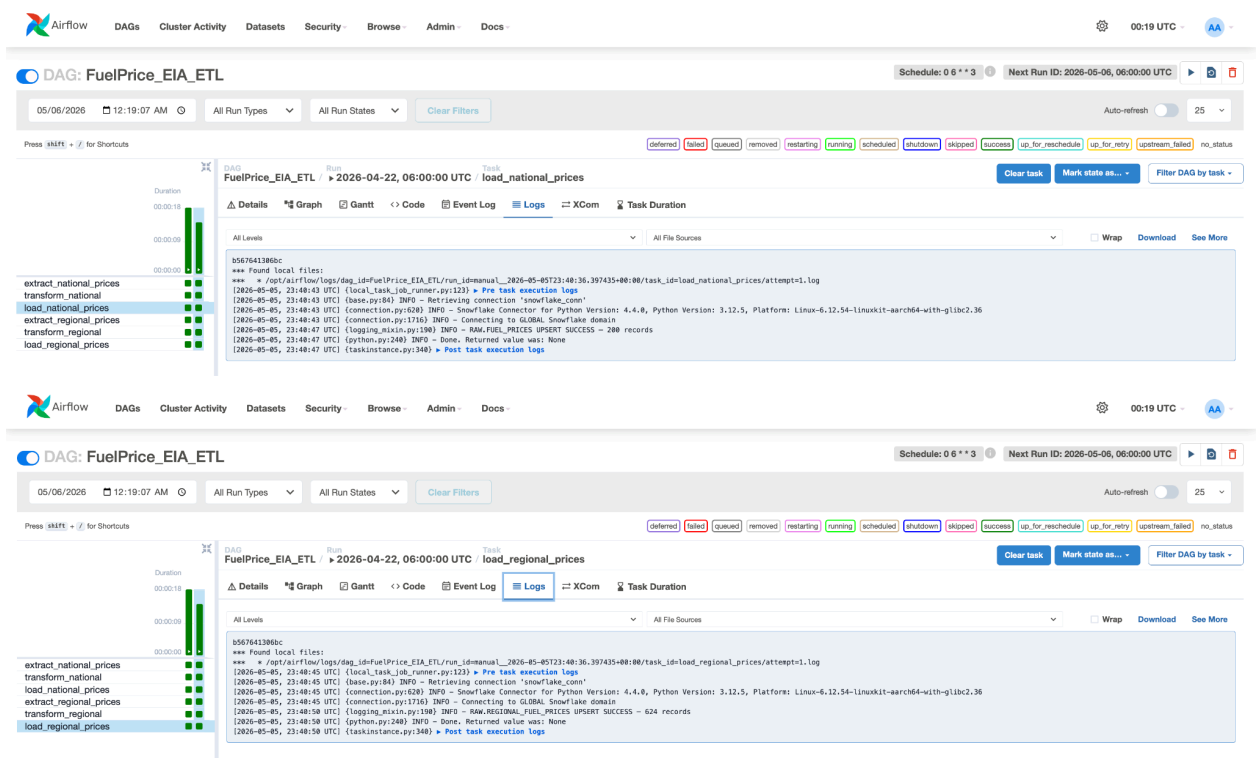


FIGURE 4: Airflow execution log output for FuelPrice_EIA_ETL

Pipeline 2 - FuelPrice_Realttime_ETL

1. Extracts energy market data from Yahoo Finance using the yfinance library.
2. Pulls weekly OHLCV data for energy-related tickers including CL=F, BZ=F, XLE, and UGA.
3. Transforms raw market data into structured records containing open, high, low, close, and volume values.

- Standardizes ticker names and aligns dates with weekly fuel price records.
- Loads transformed market data into RAW.ENERGY_MARKET_PRICES.
- Supports crude oil and fuel market comparison for downstream analytics.
- Scheduled daily at 8:00 AM UTC.

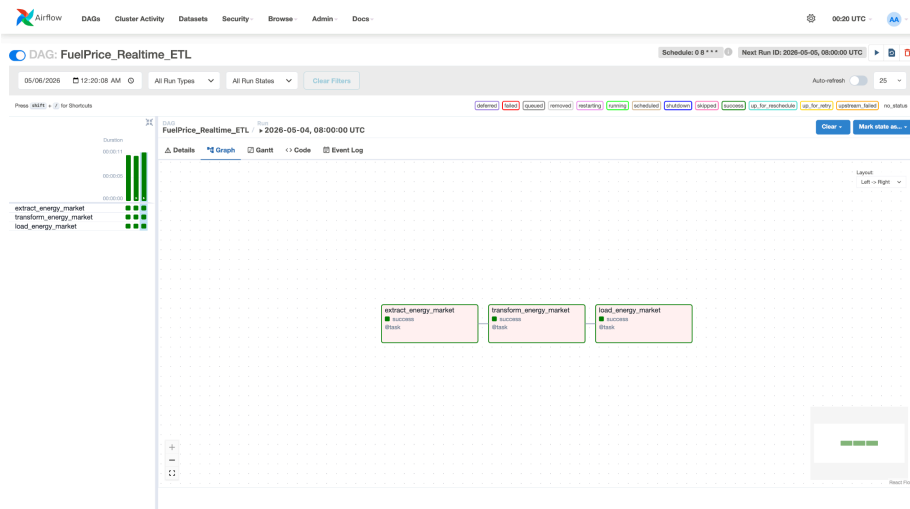


FIGURE 5: Detailed view of Pipeline 2 (FuelPrice_Realtime_ETL)

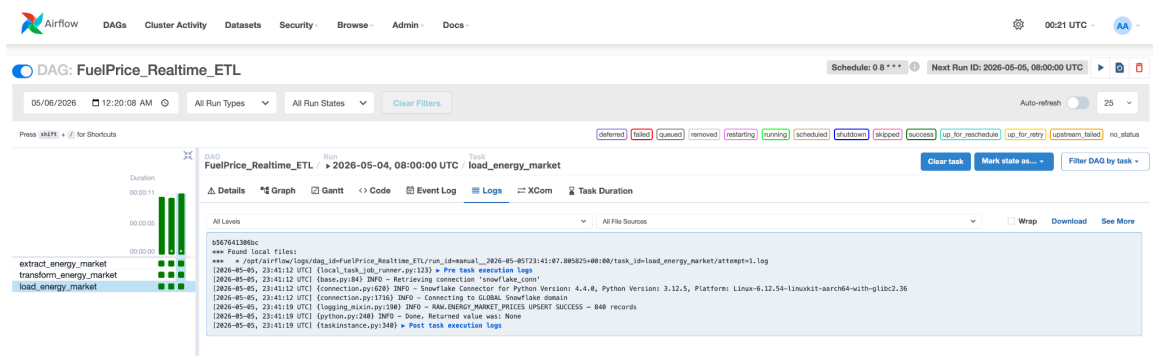


FIGURE 6: Airflow execution log output for FuelPrice_Realtime_ETL

Pipeline 3 - FuelPrice_TrainPredict

- Creates an ML training view using historical national regular gasoline prices.
- Uses WEEK_DATE as the time column and REGULAR_GASOLINE_PRICE as the target variable.
- Trains a Snowflake ML Forecast model on historical gasoline price data.
- Generates a 12-week forecast with lower and upper prediction bounds.
- Stores raw forecast results in ADHOC.FUEL_PRICE_FORECAST.
- Combines historical actual values and forecasted values into ANALYTICS.FUEL_PRICE_FINAL.
- Stores model evaluation metrics in ANALYTICS.FUEL_PRICE_MODEL_METRICS.
- Scheduled daily at 9:00 AM UTC, after the market data pipeline.

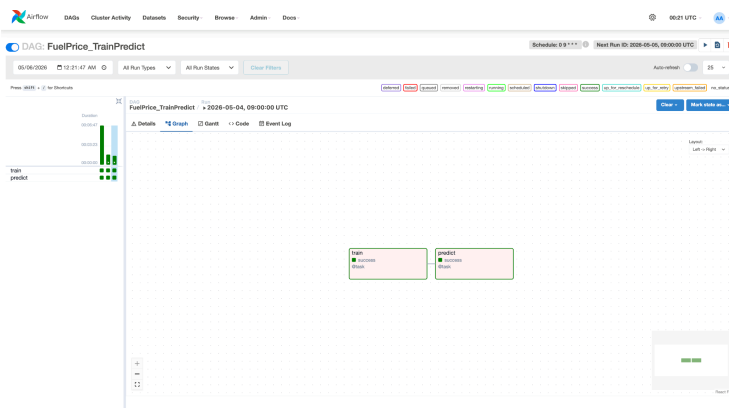
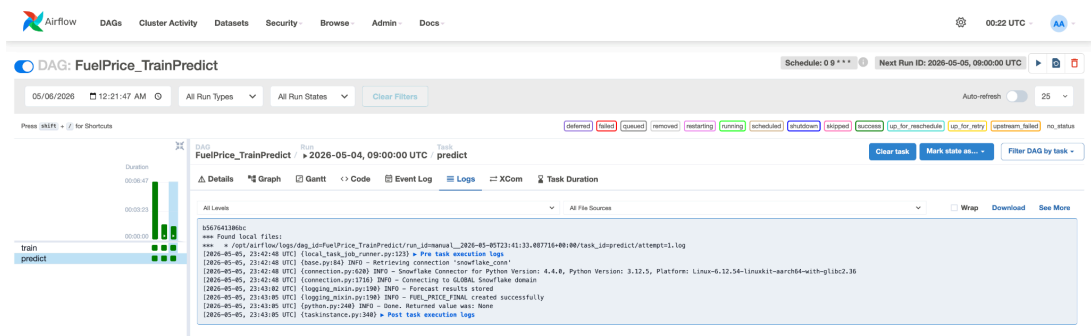


FIGURE 7: Detailed view of Pipeline 3 (FuelPrice_TrainPredict)



Airflow Connection:

Snowflake credentials are stored securely as an Airflow Connection named `snowflake_conn`. This connection allows all Airflow DAGs to access Snowflake without hardcoding credentials inside the ETL scripts. The connection uses the Snowflake account, database `USER_DB_FERRET`, warehouse `FERRET_QUERY_WH`, role `TRAINING_ROLE`, and default schema `RAW`. This configuration is used by the EIA ETL, realtime market ETL, ML forecasting, and dbt pipeline to load and transform data in Snowflake.

The screenshot shows the 'Edit Connection' form for the connection named 'snowflake_conn'. The form includes the following fields:

- Connection Id:** snowflake_conn
- Connection Type:** Snowflake
- Description:**
- Schema:** RAW
- Login:** FERRET
- Password:** (masked)
- Extra:**

```

{
  "account": "SFEDU02-EAB27764",
  "database": "USER_DB_FERRET",
  "warehouse": "FERRET_QUERY_WH",
  "role": "TRAINING_ROLE",
  "insecure_mode": false
}

```
- Account:** SFEDU02-EAB27764
- Warehouse:** FERRET_QUERY_WH
- Database:** USER_DB_FERRET

FIGURE 8: Snowflake credentials configured in the `snowflake_conn` Airflow Connection

Airflow Variable:

The EIA API key is stored as an Airflow Variable named `eia_api_key`. This variable is used by the `FuelPrice_EIA_ETL` pipeline to authenticate requests to the U.S. Energy Information Administration API and extract weekly national and regional fuel price data. Storing the API key as a variable keeps sensitive information separate from the source code.

The screenshot shows the 'Edit Variable' form for the variable named 'eia_api_key'. The form includes the following fields:

- Key:** eia_api_key
- Val:** (masked)
- Description:**

FIGURE 9: EIA API key configured as the `eia_api_key` Airflow Variable

VI. METHODOLOGY

A. Data Extraction

The project uses two primary extraction pipelines. The first pipeline, `FuelPrice_EIA_ETL`, retrieves weekly national and regional fuel price data from the EIA API. National data includes regular, midgrade, premium gasoline, and diesel prices, while regional data includes regular gasoline prices for California, West Coast, East Coast, Midwest, Gulf Coast, and Rocky Mountain regions. The second pipeline, `FuelPrice_Realtime_ETL`, extracts weekly energy market data from Yahoo Finance using the `yfinance` library for `CL=F`, `BZ=F`, `XLE`, and `UGA`.

B. Data Transformation

The transformation process converts raw API and market responses into structured records aligned with the Snowflake target schemas. National fuel price series are joined by `WEEK_DATE` to create one composite weekly record for `US_NATIONAL`. Regional fuel price records are standardized by `WEEK_DATE`, `REGION`, `FUEL_TYPE`, and `PRICE`. Yahoo Finance market data is transformed into weekly open, high, low, close, and volume records so it can be compared with retail fuel price movement.

C. Data Loading

Transformed records are loaded into Snowflake raw tables including `RAW.FUEL_PRICES`, `RAW.REGIONAL_FUEL_PRICES`, and `RAW.ENERGY_MARKET_PRICES`. Each loading task uses a `MERGE` upsert strategy to update existing records and insert new rows without creating duplicates. National fuel prices are keyed by `WEEK_DATE` and `REGION`, regional prices are keyed by `WEEK_DATE`, `REGION`, and `FUEL_TYPE`, and market prices are keyed by `WEEK_DATE` and `TICKER`.

D. dbt Transformation Models

After raw data is loaded, the `FuelPrice_DBT` pipeline runs dbt transformations to create analytics-ready models. `DBT.PRICE_MOVING_AVG` calculates rolling fuel price averages, `DBT.PRICE_VOLATILITY` computes week-over-week changes and volatility labels, `DBT.CRUDE_CORRELATION` joins gasoline prices with crude oil market data, and `DBT.REGIONAL_COMPARISON` compares each region against the U.S. national average. The `DBT.FUEL_PRICES_SNAPSHOT` model tracks historical changes using a Slowly Changing Dimension Type 2 approach.

E. Model Training

The `FuelPrice_TrainPredict` pipeline creates a clean training view called `ADHOC.FUEL_PRICE_TRAIN_VIEW` from `RAW.FUEL_PRICES`. The view selects `REGION`, `WEEK_DATE`, and `REGULAR_GASOLINE_PRICE`, filtered to the national gasoline series. A `SNOWFLAKE.ML.FORECAST` model is trained using `REGION` as the series column, `WEEK_DATE` as the timestamp column, and `REGULAR_GASOLINE_PRICE` as the prediction target.

F. Forecast Generation

The trained Snowflake ML model generates a 12-week gasoline price forecast with a 95% prediction interval. Forecast results include the predicted value, lower bound, and upper bound. The output is captured using `RESULT_SCAN(LAST_QUERY_ID())` and stored in `ADHOC.FUEL_PRICE_FORECAST` for downstream analytics.

G. Final Analytics Table

Historical and forecasted data are combined using `UNION ALL` into `ANALYTICS.FUEL_PRICE_FINAL`. Historical rows contain actual gasoline prices, while forecast rows contain forecast values and confidence bounds. A `RECORD_TYPE` column distinguishes `HISTORICAL` rows from `FORECAST` rows, making the table ready for dashboard visualization and actual-versus-forecast analysis.

VII. TABLE STRUCTURES

All tables reside in the Snowflake database `USER_DB_FERRET` using the warehouse `FERRET_QUERY_WH`. The following section describes every table maintained by the system, organized by schema.

A. Schema Overview

Schema	Purpose
RAW	Stores raw source data loaded by Airflow ETL DAGs. Contains unmodified records from EIA and Yahoo Finance.
ADHOC	Stores intermediate ML training objects and raw forecast output. Not intended for direct dashboard consumption.
ANALYTICS	Stores final forecast and model metric tables ready for BI consumption.

Schema	Purpose
DBT	Stores dbt-managed ELT transformation tables and the SCD Type 2 snapshot.

TABLE 5: Snowflake schema overview — RAW, ADHOC, ANALYTICS, and DBT schemas in USER_DB_FERRET.

B. RAW Schema

Table 1: RAW.FUEL_PRICES

Column Name	Data Type	Constraint	Description
WEEK_DATE	DATE	NOT NULL, PK	Week the price observation was recorded
REGION	STRING	NOT NULL, PK	Always US_NATIONAL for this table
REGULAR_GASOLINE_PRICE	FLOAT	—	Weekly U.S. national regular gasoline price (USD/gal)
MIDGRADE_GASOLINE_PRICE	FLOAT	—	Weekly U.S. national midgrade gasoline price (USD/gal)
PREMIUM_GASOLINE_PRICE	FLOAT	—	Weekly U.S. national premium gasoline price (USD/gal)
DIESEL_PRICE	FLOAT	—	Weekly U.S. national diesel price (USD/gal)
PRICE_UNIT	STRING	—	Unit of measurement (USD_PER_GALLON)
SOURCE	STRING	—	Data source identifier (EIA_WEEKLY)
LOAD_TS	TIMESTAMP	DEFAULT NOW()	Timestamp when the row was loaded or last updated

TABLE 6: RAW.FUEL_PRICES — Weekly U.S. national fuel prices from the EIA API. Loaded by FuelPrice_EIA_ETL.

Table 2: RAW.REGIONAL_FUEL_PRICES

Column Name	Data Type	Constraint	Description
WEEK_DATE	DATE	NOT NULL, PK	Week the price observation was recorded
REGION	STRING	NOT NULL, PK	PADD region name (e.g., EAST_COAST, CALIFORNIA)
PRICE	FLOAT	NOT NULL	Weekly regular gasoline price for the region (USD/gal)
FUEL_TYPE	STRING	—	Fuel type (REGULAR_GASOLINE)
PRICE_UNIT	STRING	—	Unit of measurement (USD_PER_GALLON)
SOURCE	STRING	—	Data source identifier (EIA_REGIONAL)
LOAD_TS	TIMESTAMP	DEFAULT NOW()	Timestamp when the row was loaded or last updated

TABLE 7: RAW.REGIONAL_FUEL_PRICES — Weekly regional regular gasoline prices by PADD region. Loaded by FuelPrice_EIA_ETL.

Table 3: RAW.ENERGY_MARKET_PRICES

Column Name	Data Type	Constraint	Description
WEEK_DATE	DATE	NOT NULL, PK	Week the market observation was recorded

Column Name	Data Type	Constraint	Description
TICKER	STRING	NOT NULL, PK	Ticker symbol (CL=F, BZ=F, XLE, UGA)
TICKER_NAME	STRING	—	Human-readable name for the ticker
OPEN_PRICE	FLOAT	—	Opening price for the week
HIGH_PRICE	FLOAT	—	Highest price during the week
LOW_PRICE	FLOAT	—	Lowest price during the week
CLOSE_PRICE	FLOAT	NOT NULL	Closing price for the week
VOLUME	BIGINT	—	Weekly trading volume
LOAD_TS	TIMESTAMP	DEFAULT NOW()	Timestamp when the row was loaded or last updated

TABLE 8: RAW.ENERGY_MARKET_PRICES — Weekly energy market OHLCV data for four tickers from Yahoo Finance. Loaded by FuelPrice_Realtime_ETL.

C. ADHOC Schema

Table / View	Created / Used By	Purpose
ADHOC.FUEL_PRICE_TRAIN_VIEW	FuelPrice_TrainPredict	Clean training view for Snowflake ML Forecast (REGION, WEEK_DATE, REGULAR_GASOLINE_PRICE)
ADHOC.FUEL_PRICE_FORECAST	FuelPrice_TrainPredict	Raw ML forecast output captured via RESULT_SCAN(LAST_QUERY_ID())

TABLE 9: ADHOC schema — Intermediate ML training and forecast objects created by FuelPrice_TrainPredict.

D. ANALYTICS Schema

Table 4: ANALYTICS.FUEL_PRICE_FINAL

Column Name	Data Type	Constraint	Description
REGION	STRING	—	Always US_NATIONAL
WEEK_DATE	DATE	—	Date of the record (historical or forecast)
ACTUAL	FLOAT	—	Historical actual gasoline price (NULL for forecast rows)
FORECAST	FLOAT	—	Forecast price (NULL for historical rows)
LOWER_BOUND	FLOAT	—	Lower 95% prediction interval (NULL for historical rows)
UPPER_BOUND	FLOAT	—	Upper 95% prediction interval (NULL for historical rows)
RECORD_TYPE	STRING	—	HISTORICAL or FORECAST

TABLE 10: ANALYTICS.FUEL_PRICE_FINAL — UNION ALL of historical actuals and 12-week ML forecast rows. Primary dashboard source.

Table 5: ANALYTICS.FUEL_PRICE_MODEL_METRICS

Column Name	Data Type	Constraint	Description
RUN_TS	TIMESTAMP	—	Timestamp of the training run
SERIES	STRING	—	Series identifier (REGION value)

Column Name	Data Type	Constraint	Description
METRIC	STRING	—	Metric name (MAE, MAPE, SMAPE, etc.)
VALUE	FLOAT	—	Metric value for that training run

TABLE 11: ANALYTICS.FUEL_PRICE_MODEL_METRICS — Snowflake ML Forecast evaluation metrics appended after each training run.

E. DBT Schema

Table 6: DBT.PRICE_MOVING_AVG

Column Name	Data Type	Description
WEEK_DATE	DATE	Date of the observation
REGION	STRING	Always US_NATIONAL
REGULAR_GASOLINE_PRICE	FLOAT	Weekly regular gasoline price (USD/gal)
MIDGRADE_GASOLINE_PRICE	FLOAT	Weekly midgrade gasoline price (USD/gal)
PREMIUM_GASOLINE_PRICE	FLOAT	Weekly premium gasoline price (USD/gal)
DIESEL_PRICE	FLOAT	Weekly diesel price (USD/gal)
REGULAR_4WK_AVG	FLOAT	4-week rolling average of regular gasoline price
DIESEL_4WK_AVG	FLOAT	4-week rolling average of diesel price
REGULAR_12WK_AVG	FLOAT	12-week rolling average of regular gasoline price
DIESEL_12WK_AVG	FLOAT	12-week rolling average of diesel price
REGULAR_52WK_AVG	FLOAT	52-week rolling average of regular gasoline price
DIESEL_52WK_AVG	FLOAT	52-week rolling average of diesel price
PREMIUM_SPREAD	FLOAT	Premium price minus regular price
DIESEL_SPREAD	FLOAT	Diesel price minus regular price

TABLE 12: DBT.PRICE_MOVING_AVG — 4-week, 12-week, and 52-week rolling averages for regular gasoline and diesel. Source: RAW.FUEL_PRICES.

Table 7: DBT.PRICE_VOLATILITY

Column Name	Data Type	Description
WEEK_DATE	DATE	Date of the observation
REGION	STRING	Always US_NATIONAL
REGULAR_GASOLINE_PRICE	FLOAT	Weekly regular gasoline price (USD/gal)
DIESEL_PRICE	FLOAT	Weekly diesel price (USD/gal)
REGULAR_WOW_CHANGE	FLOAT	Dollar change week over week (regular gasoline)
DIESEL_WOW_CHANGE	FLOAT	Dollar change week over week (diesel)
REGULAR_WOW_PCT_CHANGE	FLOAT	Percentage change week over week (regular gasoline)
HIST_AVG_REGULAR	FLOAT	All-time historical average regular gasoline price for this region

Column Name	Data Type	Description
PRICE_ANOMALY	FLOAT	Deviation from the historical average (current minus historical)
ROLLING_12WK_STDDEV	FLOAT	12-week rolling standard deviation (volatility measure)
PRICE_DIRECTION	STRING	Rising, Falling, or Stable (based on WoW change threshold)
VOLATILITY_LABEL	STRING	High, Moderate, or Low Volatility (based on 12-week STDDEV)

TABLE 13: DBT.PRICE_VOLATILITY — Week-over-week changes, rolling standard deviation, and directional/volatility labels.
Source: RAW.FUEL_PRICES.

Table 8: DBT.CRUDE_CORRELATION

Column Name	Data Type	Description
WEEK_DATE	DATE	Date of the joined observation
REGULAR_GASOLINE_PRICE	FLOAT	Weekly regular gasoline price (USD/gal)
DIESEL_PRICE	FLOAT	Weekly diesel price (USD/gal)
CRUDE_CLOSE_USD_BBL	FLOAT	WTI crude oil closing price (USD/barrel)
CRUDE_USD_PER_GALLON	FLOAT	WTI crude oil price converted to USD/gallon ($\div 42$)
CRUDE_LAG2_BBL	FLOAT	WTI crude oil price 2 weeks prior (USD/barrel)
CRUDE_LAG2_GAL	FLOAT	WTI crude oil price 2 weeks prior converted to USD/gallon
PUMP_TO_CRUDE_MARGIN	FLOAT	Retail gasoline price minus lagged crude cost per gallon
MARGIN_CATEGORY	STRING	Wide Margin, Normal Margin, or Tight Margin

TABLE 14: DBT.CRUDE_CORRELATION — Retail gasoline prices joined with WTI crude oil futures with a 2-week lag. Sources: RAW.FUEL_PRICES + RAW.ENERGY_MARKET_PRICES.

Table 9: DBT.REGIONAL_COMPARISON

Column Name	Data Type	Description
WEEK_DATE	DATE	Date of the observation
REGION	STRING	PADD region name (EAST_COAST, MIDWEST, etc.)
REGIONAL_PRICE	FLOAT	Weekly regular gasoline price for the region (USD/gal)
NATIONAL_PRICE	FLOAT	U.S. national weekly regular gasoline average (USD/gal)
PRICE_SPREAD	FLOAT	Regional price minus national price
PRICE_SPREAD_PCT	FLOAT	Percentage spread relative to national average
REGIONAL_4WK_AVG	FLOAT	4-week rolling average of regional price
MAX_SPREAD_SEEN	FLOAT	Running maximum spread seen so far for this region
PRICE_CATEGORY	STRING	Significantly Above / Above / Near / Below / Significantly Below Average

TABLE 15: DBT.REGIONAL_COMPARISON — Regional regular gasoline prices compared against the U.S. national average.
Sources: RAW.FUEL_PRICES + RAW.REGIONAL_FUEL_PRICES.

Table 10: DBT.FUEL_PRICES_SNAPSHOT (SCD Type 2)

Column Name	Data Type	Description
WEEK_DATE	DATE	Week of the fuel price record
REGION	STRING	Region (US_NATIONAL)
REGULAR_GASOLINE_PRICE	FLOAT	Regular gasoline price at the time of the snapshot
MIDGRADE_GASOLINE_PRICE	FLOAT	Midgrade gasoline price at the time of the snapshot
PREMIUM_GASOLINE_PRICE	FLOAT	Premium gasoline price at the time of the snapshot
DIESEL_PRICE	FLOAT	Diesel price at the time of the snapshot
PRICE_UNIT	STRING	Unit of measurement
SOURCE	STRING	Data source identifier
LOAD_TS	TIMESTAMP	Original load timestamp
DBT_SCD_ID	STRING	dbt-generated unique row identifier
DBT_UPDATED_AT	TIMESTAMP	Timestamp of the most recent update
DBT_VALID_FROM	TIMESTAMP	Start of validity for this row version
DBT_VALID_TO	TIMESTAMP	End of validity (NULL if current record)
DBT_IS_CURRENT_RECORD	BOOLEAN	TRUE if this is the current active record for that key

TABLE 16: DBT.FUEL_PRICES_SNAPSHOT — SCD Type 2 dbt snapshot tracking all historical changes to RAW.FUEL_PRICES over time.

VIII. IMPLEMENTATION

A. Technologies Used

- Apache Airflow: pipeline orchestration via Docker Compose (LocalExecutor)
- Snowflake: cloud data warehouse, Snowflake ML Forecast, DBT schema storage
- EIA API: official U.S. government weekly fuel price data source
- Yahoo Finance / yfinance: free Python library for energy market OHLCV data
- Python: ETL DAG logic, data transformation, and API integration
- dbt (data build tool): ELT transformation models, schema tests, and SCD Type 2 snapshots
- Preset: BI dashboard and interactive visualization layer
- Docker / Docker Compose: containerized local Airflow deployment

B. Airflow Variables and Connections

- Variable `eia_api_key`: Stores the EIA API key securely as a plain-text Airflow Variable. The `FuelPrice_EIA_ETL` DAG retrieves it at runtime using `Variable.get('eia_api_key')`, keeping the key entirely out of the source code.
- Connection `snowflake_conn`: Stores all Snowflake credentials securely. All four pipelines retrieve the connection via `SnowflakeHook(snowflake_conn_id='snowflake_conn')` or by reading environment variables injected from the connection at runtime. The connection stores the account (`SFEDU02-EAB27764`), database (`USER_DB_FERRET`), warehouse (`FERRET_QUERY_WH`), role (`TRAINING_ROLE`), login (`FERRET`), and password. No credentials appear anywhere in the DAG source files or dbt configuration files.
- For the `FuelPrice_DBT` DAG specifically, connection parameters are injected as environment variables (`DBT_TYPE`, `DBT_ACCOUNT`, `DBT_USER`, `DBT_PASSWORD`, `DBT_DATABASE`, `DBT_SCHEMA`, `DBT_WAREHOUSE`, `DBT_ROLE`) using Airflow's Jinja templating inside the `BashOperator` env dictionary. The `profiles.yml` file reads these environment variables using dbt's `env_var()` function, making the profile completely free of hardcoded secrets.

C. ETL DAG - FuelPrice_EIA_ETL

The FuelPrice_EIA_ETL DAG is scheduled at 6:00 AM UTC every Wednesday (cron: 0 6 * * 3). It executes two parallel branches, each with three sequential tasks: extract, transform, and load. The national branch loads four fuel type series into RAW.FUEL_PRICES; the regional branch loads six PADD region series into RAW.REGIONAL_FUEL_PRICES. Both load tasks use a temporary staging table with MERGE logic to guarantee idempotent upserts. Re-running the DAG on the same data updates existing rows and inserts new ones, but never creates duplicates.

D. ETL DAG - FuelPrice_Realtime_ETL

The FuelPrice_Realtime_ETL DAG is scheduled daily at 8:00 AM UTC (cron: 0 8 * * *). It executes a single linear pipeline of three tasks. The extract task calls yfinance.Ticker.history() for each of the four tickers with a 4-year lookback and weekly interval. The transform task flattens the dataframe rows into typed tuples. The load task MERGE upserts into RAW.ENERGY_MARKET_PRICES keyed on (WEEK_DATE, TICKER). All operations are wrapped in a SQL transaction with try/except/rollback.

E. ML DAG - FuelPrice_TrainPredict

The FuelPrice_TrainPredict DAG is scheduled daily at 9:00 AM UTC. It reads from RAW.FUEL_PRICES, which is refreshed weekly by the FuelPrice_EIA_ETL DAG. It contains two sequential tasks: train and predict. The train task builds the Snowflake ML Forecast model using CREATE OR REPLACE SNOWFLAKE.ML.FORECAST and appends evaluation metrics to the metrics table. The predict task calls the model's FORECAST method for 12 periods, captures the output using RESULT_SCAN(LAST_QUERY_ID()), and builds the ANALYTICS.FUEL_PRICE_FINAL UNION ALL table.

F. dbt DAG - FuelPrice_DBT

The FuelPrice_DBT DAG is scheduled daily at 10:00 AM UTC. It uses BashOperator to invoke dbt commands using the profiles directory and project directory both set to /opt/airflow/dbt, where the dbt project files are volume-mounted from the host machine. Three tasks run sequentially: dbt_run builds all four transformation models, dbt_test validates all model outputs against schema test definitions, and dbt_snapshot updates the fuel_prices_snapshot.

G. dbt Testing

Tests are defined in schema.yml covering all four dbt models and all three RAW source tables. The dbt testing framework validates data quality automatically after every model build. The following categories of tests are enforced:

Model / Source	Test Type	Columns Tested
RAW source (fuel_prices)	not_null	WEEK_DATE, REGION, REGULAR_GASOLINE_PRICE, SOURCE
RAW source (fuel_prices)	accepted_values	REGION must be 'US_NATIONAL'; SOURCE must be 'EIA_WEEKLY'
RAW source (regional_fuel_prices)	not_null	WEEK_DATE, REGION, PRICE
RAW source (regional_fuel_prices)	accepted_values	REGION: 6 PADD values; FUEL_TYPE: REGULAR_GASOLINE
RAW source (energy_market_prices)	not_null	WEEK_DATE, TICKER, CLOSE_PRICE
RAW source (energy_market_prices)	accepted_values	TICKER: CL=F, BZ=F, XLE, UGA
price_moving_avg	not_null	WEEK_DATE, REGION, REGULAR_GASOLINE_PRICE, REGULAR_4WK_AVG, REGULAR_12WK_AVG
price_volatility	not_null	WEEK_DATE, REGION, REGULAR_GASOLINE_PRICE
price_volatility	accepted_values	PRICE_DIRECTION: Rising, Falling, Stable; VOLATILITY_LABEL: 3 tiers

Model / Source	Test Type	Columns Tested
crude_correlation	not_null	WEEK_DATE, REGULAR_GASOLINE_PRICE
crude_correlation	accepted_values	MARGIN_CATEGORY: Wide Margin, Normal Margin, Tight Margin
regional_comparison	not_null	WEEK_DATE, REGION, REGIONAL_PRICE
regional_comparison	accepted_values	PRICE_CATEGORY: 5 label values

TABLE 17: dbt schema test coverage across all models and RAW source tables.

H. Key Design Decisions

MERGE over INSERT: All ETL load tasks use Snowflake MERGE with temporary staging tables rather than direct INSERT statements. This guarantees that re-running any DAG — due to a retry, a manual trigger, or a scheduled replay produces exactly the same result without duplicate records.

Staggered DAG scheduling: The one-hour offsets between EIA ETL (6:00 AM Wed), Realtime ETL (8:00 AM daily), TrainPredict (9:00 AM daily), and DBT (10:00 AM daily) ensure each pipeline always runs after its upstream dependencies have completed, without requiring complex cross-DAG dependencies or Airflow sensors.

Credential injection via Jinja templating: The FuelPrice_DBT DAG passes all Snowflake connection parameters as environment variables using Airflow's Jinja template syntax inside the BashOperator env dictionary. The profiles.yml file contains no secrets. This keeps the single Airflow Connection as the source of truth for credentials across all four pipelines.

ON_ERROR: SKIP in ML training: The SNOWFLAKE.ML.FORECAST model is configured with ON_ERROR: SKIP, which prevents a data quality issue in one series from blocking forecast generation for all other series. This improves overall pipeline resilience without requiring manual intervention.

RESULT_SCAN pattern: Snowflake ML Forecast output cannot be written directly to a table using a simple CREATE TABLE AS SELECT. The forecast result must be captured immediately using RESULT_SCAN(LAST_QUERY_ID()) in the same session, immediately after the CALL statement. This non-obvious requirement was identified and handled during implementation.

2-week crude oil lag in crude_correlation: The crude_correlation dbt model applies a 2-week LAG to WTI crude oil prices before joining them with retail gasoline prices. This lag is based on the well-established pattern that crude oil price changes typically take 1–3 weeks to flow through refining, distribution, and retail pricing before appearing at the pump. The lag model makes the correlation analysis more analytically accurate than a direct same-week join.

I. Source Code Repository

All Python DAG files and SQL queries for this project are publicly available on GitHub: <https://github.com/Kshitija-Shinde9/End-to-End-Dbt-Airflow-Snowflake-Data-Pipeline-Fuel-Price-Analysis>

IX. RESULTS

A. Pipeline Execution

All four Airflow DAGs executed successfully across scheduled and manual runs with no failures or retries required. The FuelPrice_EIA_ETL DAG completed the national and regional EIA loads within expected runtime. The FuelPrice_Realtime_ETL DAG successfully fetched 4 years of weekly OHLCV data for all four tickers and loaded them into RAW.ENERGY_MARKET_PRICES. The FuelPrice_TrainPredict DAG completed model training, forecast generation, and FUEL_PRICE_FINAL table creation within a few minutes. The FuelPrice_DBT DAG built all four models, passed all schema tests, and updated the snapshot successfully.

B. Snowflake Tables Loaded

After all pipeline runs, the Snowflake warehouse contains data across all four schemas. RAW.FUEL_PRICES holds the latest 200 weekly national records (approximately 4 years of data). RAW.REGIONAL_FUEL_PRICES holds the latest 104 weekly records per PADD region (approximately 2 years per region). RAW.ENERGY_MARKET_PRICES holds up to 4 years of weekly OHLCV data for all four tickers. The DBT schema contains four analytics-ready transformation tables and one SCD Type 2 snapshot table.

Sample verification queries for Snowflake:

```
SELECT * FROM USER_DB_FERRET.RAW.FUEL_PRICES;
```

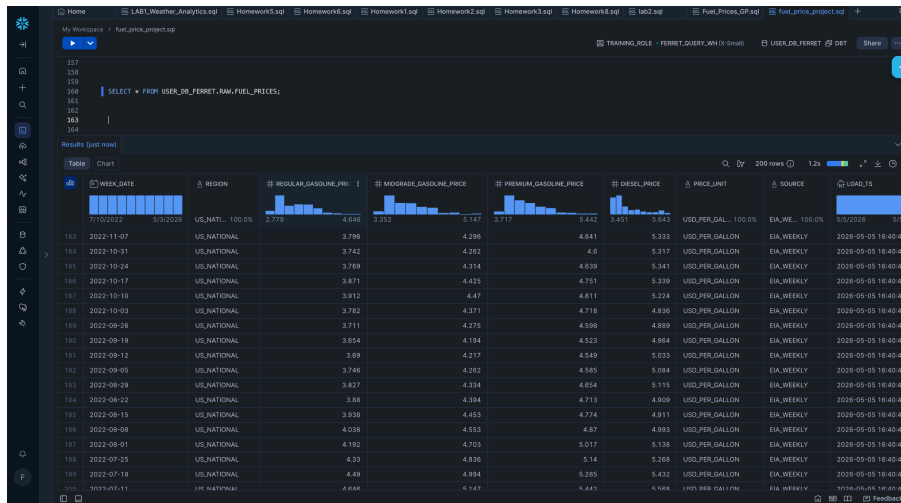



FIGURE 10: Snowflake output showing populated *RAW.FUEL_PRICES*
SELECT * FROM USER_DB_FERRET.RAW.REGIONAL_FUEL_PRICES;

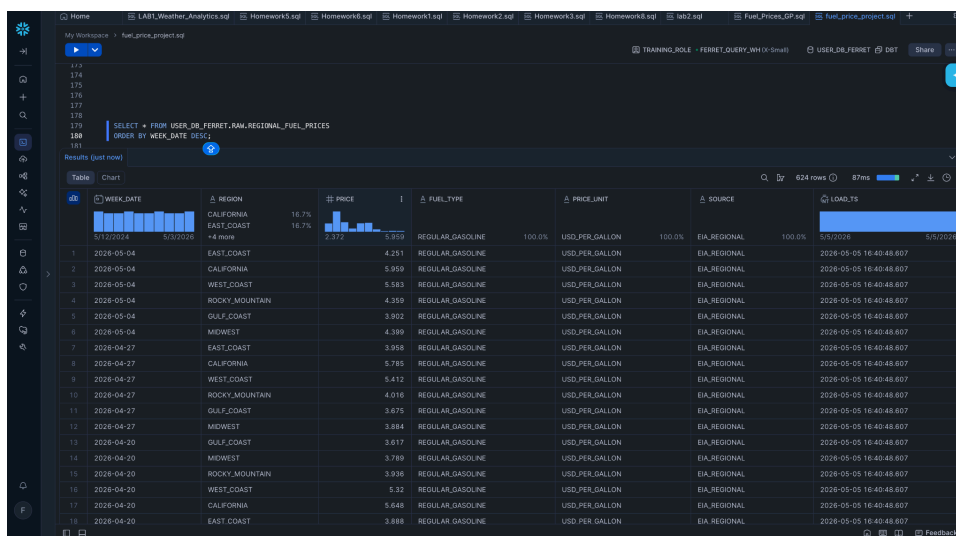


FIGURE 11: Snowflake output showing populated *RAW.REGIONAL_FUEL_PRICES*

SELECT * FROM USER_DB_FERRET.RAW.ENERGY_MARKET_PRICES;

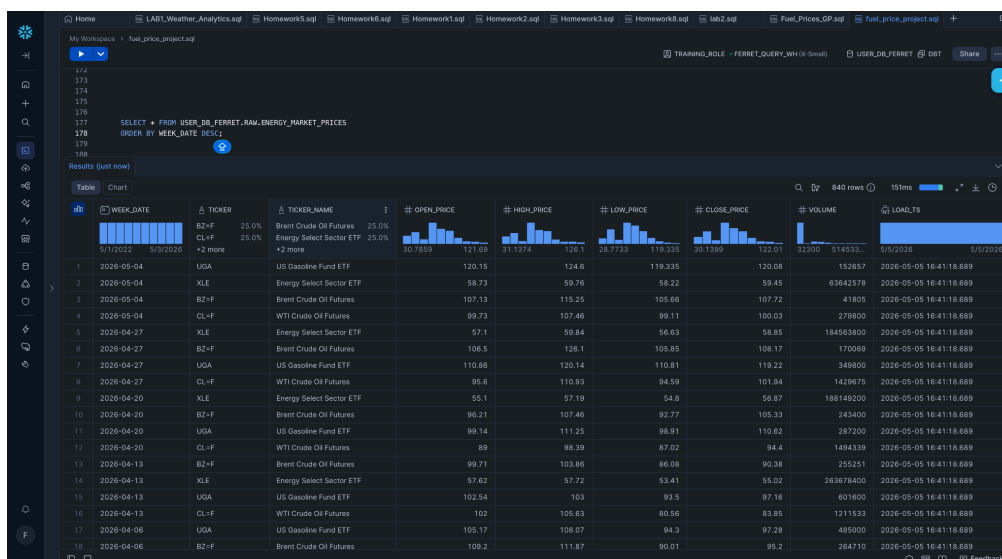


FIGURE 12: Snowflake output showing populated *RAW.ENERGY_MARKET_PRICES*

**SELECT TICKER, COUNT(*) FROM USER_DB_FERRET.RAW.ENERGY_MARKET_PRICES
 GROUP BY TICKER ORDER BY TICKER;**

```

314
315 SELECT TICKER, COUNT(*) FROM USER_DB_FERRET.RAW.ENERGY_MARKET_PRICES
316 GROUP BY TICKER ORDER BY TICKER;
317

```

Results (just now)

TICKER	# COUNT(*)
BZ=F	210
CL=F	210
UGA	210
XLE	210

FIGURE 13: Snowflake output showing row count by ticker for RAW.ENERGY_MARKET_PRICES

SELECT * FROM USER_DB_FERRET.DBT.PRICE_MOVING_AVG;

```

322 SELECT * FROM USER_DB_FERRET.DBT.PRICE_MOVING_AVG;
323
324

```

Results (just now)

WEEK_DATE	REGION	# REGULAR_GASOLINE_PRICE	# MIDGRADE_GASOLINE_PRICE	# PREMIUM_GASOLINE_PRICE	# DIESEL_PRICE	# REGULAR_4WK_AVG	# DIESEL_4WK_AVG	# REGULAR_12WK_AVG	# DIESEL_12WK_AVG
2026-03-16	US_NATIONAL	3.72	4.343	4.895	5.071	3.2935	4.409	2.9927	3.81
2026-03-23	US_NATIONAL	3.961	4.579	4.943	5.375	3.5495	4.8005	3.0885	4.0
2026-03-30	US_NATIONAL	3.99	4.612	4.984	5.401	3.7932	5.1765	3.188	4.11
2026-04-06	US_NATIONAL	4.12	4.727	5.102	5.643	3.9477	5.3725	3.2997	4.31
2026-04-13	US_NATIONAL	4.123	4.715	5.088	5.608	4.0485	5.5088	3.4095	4.51
2026-04-20	US_NATIONAL	4.044	4.655	5.024	5.403	4.0692	5.5138	3.5087	4.61
2026-04-27	US_NATIONAL	4.123	4.737	5.097	5.351	4.1025	5.5013	3.6134	4.81
2026-05-04	US_NATIONAL	4.452	5.025	5.402	5.64	4.1855	5.5005	3.7426	4.91

FIGURE 14: Snowflake output showing populated DBT.PRICE_MOVING_AVG

SELECT * FROM USER_DB_FERRET.DBT.PRICE_VOLATILITY;

```

325 SELECT * FROM USER_DB_FERRET.DBT.PRICE_VOLATILITY;
326
327
328

```

Results (just now)

WEEK_DATE	REGION	# REGULAR_GASOLINE_PRICE	# DIESEL_PRICE	# REGULAR_WOW_CHANGE	# DIESEL_WOW_CHANGE	# REGULAR_WOW_PCT_CHANGE	# HIST_AVG_REGULAR	# PRICE_ANOMALY
2022-07-11	US_NATIONAL	4.646	5.568					
2022-07-18	US_NATIONAL	4.49	5.432	-0.156	-0.138	-3.3577	3.3793	1.1101
2022-07-25	US_NATIONAL	4.33	5.268	-0.16	-0.164	-3.5635	3.3793	0.9501
2022-08-01	US_NATIONAL	4.192	5.138	-0.138	-0.13	-3.1871	3.3793	0.8121
2022-08-08	US_NATIONAL	4.038	4.993	-0.154	-0.145	-3.6737	3.3793	0.6581
2022-08-15	US_NATIONAL	3.938	4.911	-0.1	-0.082	-2.4765	3.3793	0.5581
2022-08-22	US_NATIONAL	3.88	4.909	-0.058	-0.002	-1.4728	3.3793	0.5001
2022-08-29	US_NATIONAL	3.827	5.115	-0.053	0.206	-1.368	3.3793	0.4471

FIGURE 15: Snowflake output showing populated DBT.PRICE_VOLATILITY

SELECT *FROM USER_DB_FERRET.DBT.CRUDE_CORRELATION;

```

329 SELECT * FROM USER_DB_FERRET.DBT.CRUDE_CORRELATION;
330

```

Results (just now)

WEEK_DATE	# REGULAR_GASOLINE_PRICE	# DIESEL_PRICE	# CRUDE_CLOSE_USD_BBL	# CRUDE_USD_PER_GALLON	# CRUDE_LAG2_BBL	# CRUDE_LAG2_GAL	# PUMP_TO_CRUDE_MARGIN	MARGIN_CATEG
2022-07-11	4.646	5.568	97.59	2.3236	108.43	2.5817	2.0643	Wide Margin
2022-07-18	4.49	5.432	94.7	2.2548	104.79	2.495	1.995	Wide Margin
2022-07-25	4.33	5.268	98.62	2.3481	97.59	2.3236	2.0084	Wide Margin
2022-08-01	4.192	5.138	89.01	2.1193	94.7	2.2548	1.9372	Wide Margin
2022-08-08	4.038	4.993	92.09	2.1926	98.62	2.3481	1.6899	Wide Margin
2022-08-15	3.938	4.911	90.77	2.1612	89.01	2.1193	1.8187	Wide Margin
2022-08-22	3.88	4.909	93.06	2.2157	92.09	2.1926	1.6874	Wide Margin
2022-08-29	3.827	5.115	86.87	2.0683	90.77	2.1612	1.6658	Wide Margin

FIGURE 16: Snowflake output showing populated DBT.CRUDE_CORRELATION

SELECT * FROM USER_DB_FERRET.DBT.REGIONAL_COMPARISON;

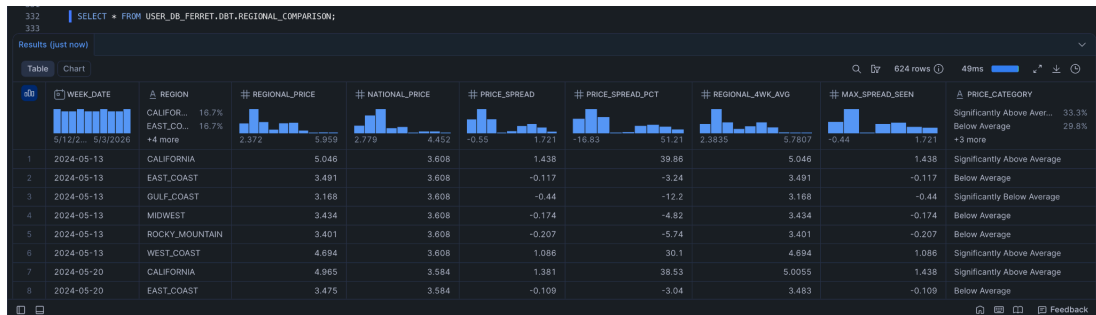


FIGURE 17: Snowflake output showing populated DBT.REGIONAL_COMPARISON

SELECT * FROM USER_DB_FERRET.ANALYTICS.FUEL_PRICE_FINAL;



FIGURE 18a : Snowflake output showing populated ANALYTICS.FUEL_PRICE_FINAL



FIGURE 18b : Snowflake output showing populated ANALYTICS.FUEL_PRICE_FINAL

C. dbt Run Results

All four dbt models built successfully with PASS=4, WARN=0, ERROR=0. The model build order is resolved automatically by dbt's internal dependency graph based on source() references in schema.yml. The four models built are: price_moving_avg, price_volatility, crude_correlation, and regional_comparison. All models are materialized as tables in the DBT schema of USER_DB_FERRET.

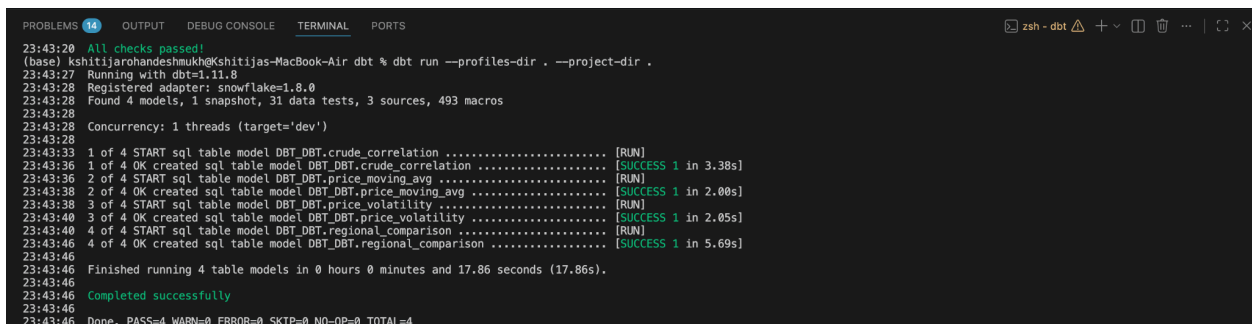


FIGURE 19: dbt run output showing 4/4 models SUCCESS

D. dbt Test Results

The dbt test run executed all schema tests defined in schema.yml across all four models and the three RAW source tables. Tests covered not_null constraints on all key columns, accepted_values validations for categorical label columns

(PRICE_DIRECTION, VOLATILITY_LABEL, MARGIN_CATEGORY, PRICE_CATEGORY), and accepted_values validations for lookup columns (REGION, TICKER, SOURCE, FUEL_TYPE). All tests passed with WARN=0 and ERROR=0.

```
PROBLEMS 14 OUTPUT DEBUG CONSOLE TERMINAL PORTS zsh - dbt + v [] [] ... | [] [] ×

23:43:46 Done. PASS=4 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=4
(base) kshiti@jarohandeshmukh@Kshiti@MacBook-Air: dbt % dbt test --profiles-dir . --project-dir .
23:43:52 Running with dbt=1.11.8
23:43:54 Registered adapter: snowflake=1.8.0
23:43:54 Found 4 models, 1 snapshot, 31 data tests, 3 sources, 493 macros
23:43:54
23:43:54 Concurrency: 1 threads (target='dev')
23:43:54
23:44:00 1 of 31 START test accepted_values_crude_correlation_MARGIN_CATEGORY_Wide_Margin_Normal_Margin_Tight_Margin [RUN]
23:44:02 1 of 31 PASS accepted_values_crude_correlation_MARGIN_CATEGORY_Wide_Margin_Normal_Margin_Tight_Margin [PASS in 1.85s]
23:44:02 2 of 31 START test accepted_values_price_volatility_PRICE_DIRECTION_Rising_Falling_Stable [RUN]
23:44:04 2 of 31 PASS accepted_values_price_volatility_PRICE_DIRECTION_Rising_Falling_Stable [PASS in 1.67s]
23:44:04 3 of 31 START test accepted_values_price_volatility_VOLATILITY_LABEL_High_Volatility_Moderate_Volatility_Low_Volatility [RUN]
23:44:06 3 of 31 PASS accepted_values_price_volatility_VOLATILITY_LABEL_High_Volatility_Moderate_Volatility_Low_Volatility [PASS in 2.49s]
23:44:06 4 of 31 START test accepted_values_regional_comparison_PRICE_CATEGORY_Significantly_Above_Average_Above_Average_Near_National_Average_Below_Average_Significantly_Below_Average [RUN]
23:44:08 4 of 31 PASS accepted_values_regional_comparison_PRICE_CATEGORY_Significantly_Above_Average_Above_Average_Near_National_Average_Below_Average_Significantly_Below_Average [PASS in 2.13s]
23:44:08 5 of 31 START test not_null_crude_correlation_REGULAR_GASOLINE_PRICE [RUN]
23:44:11 5 of 31 PASS not_null_crude_correlation_REGULAR_GASOLINE_PRICE [PASS in 2.42s]
23:44:11 6 of 31 START test not_null_crude_correlation_WEEK_DATE [RUN]
23:44:11 6 of 31 PASS not_null_crude_correlation_WEEK_DATE [PASS in 0.76s]
23:44:11 7 of 31 START test not_null_price_moving_avg_REGION [RUN]
23:44:13 7 of 31 PASS not_null_price_moving_avg_REGION [PASS in 1.29s]
23:44:13 8 of 31 START test not_null_price_moving_avg_REGULAR_12WK_AVG [RUN]
23:44:14 8 of 31 PASS not_null_price_moving_avg_REGULAR_12WK_AVG [PASS in 0.90s]
23:44:14 9 of 31 START test not_null_price_moving_avg_REGULAR_4WK_AVG [RUN]
23:44:15 9 of 31 PASS not_null_price_moving_avg_REGULAR_4WK_AVG [PASS in 1.52s]
23:44:15 10 of 31 START test not_null_price_moving_avg_REGULAR_GASOLINE_PRICE [RUN]
23:44:16 10 of 31 PASS not_null_price_moving_avg_REGULAR_GASOLINE_PRICE [PASS in 1.03s]
23:44:16 11 of 31 START test not_null_price_moving_avg_WEEK_DATE [RUN]
23:44:17 11 of 31 PASS not_null_price_moving_avg_WEEK_DATE [PASS in 0.82s]
23:44:17 12 of 31 START test not_null_price_volatility_REGION [RUN]
23:44:18 12 of 31 PASS not_null_price_volatility_REGION [PASS in 1.38s]
23:44:18 13 of 31 START test not_null_price_volatility_REGULAR_GASOLINE_PRICE [RUN]
23:44:21 13 of 31 PASS not_null_price_volatility_REGULAR_GASOLINE_PRICE [PASS in 3.18s]
23:44:21 14 of 31 START test not_null_price_volatility_WEEK_DATE [RUN]
23:44:25 14 of 31 PASS not_null_price_volatility_WEEK_DATE [PASS in 3.29s]
23:44:25 15 of 31 START test not_null_regional_comparison_REGION [RUN]
23:44:26 15 of 31 PASS not_null_regional_comparison_REGION [PASS in 1.09s]
23:44:26 16 of 31 START test not_null_regional_comparison_REGIONAL_PRICE [RUN]
23:44:27 16 of 31 PASS not_null_regional_comparison_REGIONAL_PRICE [PASS in 1.04s]
23:44:27 17 of 31 START test not_null_regional_comparison_WEEK_DATE [RUN]
23:44:28 17 of 31 PASS not_null_regional_comparison_WEEK_DATE [PASS in 1.03s]
23:44:28 18 of 31 START test source_accepted_values_raw_energy_market_prices_TICKER_CL_F_BZ_F_XLE_UGA [RUN]
23:44:29 18 of 31 PASS source_accepted_values_raw_energy_market_prices_TICKER_CL_F_BZ_F_XLE_UGA [PASS in 1.06s]
23:44:29 19 of 31 START test source_accepted_values_raw_fuel_prices_REGION_US_NATIONAL [RUN]
23:44:31 19 of 31 PASS source_accepted_values_raw_fuel_prices_REGION_US_NATIONAL [PASS in 1.89s]
23:44:31 20 of 31 START test source_accepted_values_raw_fuel_prices_SOURCE_EIA_WEEKLY [RUN]
23:44:32 20 of 31 PASS source_accepted_values_raw_fuel_prices_SOURCE_EIA_WEEKLY [PASS in 0.96s]
23:44:32 21 of 31 START test source_accepted_values_raw_regional_fuel_prices_FUEL_TYPE_REGULAR_GASOLINE [RUN]
23:44:34 21 of 31 PASS source_accepted_values_raw_regional_fuel_prices_FUEL_TYPE_REGULAR_GASOLINE [PASS in 2.13s]
23:44:34 22 of 31 START test source_accepted_values_raw_regional_fuel_prices_REGION_EAST_COAST_MIDWEST_GULF_COAST_ROCKY_MOUNTAIN_WEST_COAST_CALIFORNIA [RUN]
23:44:35 22 of 31 PASS source_accepted_values_raw_regional_fuel_prices_REGION_EAST_COAST_MIDWEST_GULF_COAST_ROCKY_MOUNTAIN_WEST_COAST_CALIFORNIA [PASS in 0.98s]
23:44:35 23 of 31 START test source_not_null_raw_energy_market_prices_CLOSE_PRICE [RUN]
23:44:36 23 of 31 PASS source_not_null_raw_energy_market_prices_CLOSE_PRICE [PASS in 0.96s]
23:44:36 24 of 31 START test source_not_null_raw_energy_market_prices_TICKER [RUN]
23:44:37 24 of 31 PASS source_not_null_raw_energy_market_prices_TICKER [PASS in 1.02s]
23:44:37 25 of 31 START test source_not_null_raw_energy_market_prices_WEEK_DATE [RUN]
23:44:38 25 of 31 PASS source_not_null_raw_energy_market_prices_WEEK_DATE [PASS in 1.14s]
23:44:38 26 of 31 START test source_not_null_raw_fuel_prices_REGION [RUN]
23:44:39 26 of 31 PASS source_not_null_raw_fuel_prices_REGION [PASS in 0.67s]
23:44:39 27 of 31 START test source_not_null_raw_fuel_prices_REGULAR_GASOLINE_PRICE [RUN]
23:44:40 27 of 31 PASS source_not_null_raw_fuel_prices_REGULAR_GASOLINE_PRICE [PASS in 0.94s]
23:44:40 28 of 31 START test source_not_null_raw_fuel_prices_WEEK_DATE [RUN]
23:44:41 28 of 31 PASS source_not_null_raw_fuel_prices_WEEK_DATE [PASS in 1.05s]
23:44:41 29 of 31 START test source_not_null_raw_regional_fuel_prices_PRICE [RUN]
23:44:42 29 of 31 PASS source_not_null_raw_regional_fuel_prices_PRICE [PASS in 0.86s]
23:44:42 30 of 31 START test source_not_null_raw_regional_fuel_prices_REGION [RUN]
23:44:43 30 of 31 PASS source_not_null_raw_regional_fuel_prices_REGION [PASS in 1.49s]
23:44:43 31 of 31 START test source_not_null_raw_regional_fuel_prices_WEEK_DATE [RUN]
23:44:44 31 of 31 PASS source_not_null_raw_regional_fuel_prices_WEEK_DATE [PASS in 0.87s]
23:44:44
23:44:44 Finished running 31 data tests in 0 hours 0 minutes and 50.06 seconds (50.06s).
23:44:44 Completed successfully
23:44:44
23:44:44 Done. PASS=31 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=31
```

FIGURE 20 a: dbt test showing 31/31 PASS

```
PROBLEMS 14 OUTPUT DEBUG CONSOLE TERMINAL PORTS zsh - dbt + v [] [] ... | [] [] ×

23:44:06 4 of 31 START test accepted_values_regional_comparison_PRICE_CATEGORY_Significantly_Above_Average_Above_Average_Near_National_Average_Below_Average_Significantly_Below_Average [RUN]
23:44:08 4 of 31 PASS accepted_values_regional_comparison_PRICE_CATEGORY_Significantly_Above_Average_Above_Average_Near_National_Average_Below_Average_Significantly_Below_Average [PASS in 2.13s]
23:44:08 5 of 31 START test not_null_crude_correlation_REGULAR_GASOLINE_PRICE [RUN]
23:44:11 5 of 31 PASS not_null_crude_correlation_REGULAR_GASOLINE_PRICE [PASS in 2.42s]
23:44:11 6 of 31 START test not_null_crude_correlation_WEEK_DATE [RUN]
23:44:11 6 of 31 PASS not_null_crude_correlation_WEEK_DATE [PASS in 0.76s]
23:44:11 7 of 31 START test not_null_price_moving_avg_REGION [RUN]
23:44:13 7 of 31 PASS not_null_price_moving_avg_REGION [PASS in 1.29s]
23:44:13 8 of 31 START test not_null_price_moving_avg_REGULAR_12WK_AVG [RUN]
23:44:14 8 of 31 PASS not_null_price_moving_avg_REGULAR_12WK_AVG [PASS in 0.90s]
23:44:14 9 of 31 START test not_null_price_moving_avg_REGULAR_4WK_AVG [RUN]
23:44:15 9 of 31 PASS not_null_price_moving_avg_REGULAR_4WK_AVG [PASS in 1.52s]
23:44:15 10 of 31 START test not_null_price_moving_avg_REGULAR_GASOLINE_PRICE [RUN]
23:44:16 10 of 31 PASS not_null_price_moving_avg_REGULAR_GASOLINE_PRICE [PASS in 1.03s]
23:44:16 11 of 31 START test not_null_price_moving_avg_WEEK_DATE [RUN]
23:44:17 11 of 31 PASS not_null_price_moving_avg_WEEK_DATE [PASS in 0.82s]
23:44:17 12 of 31 START test not_null_price_volatility_REGION [RUN]
23:44:18 12 of 31 PASS not_null_price_volatility_REGION [PASS in 1.38s]
23:44:18 13 of 31 START test not_null_price_volatility_REGULAR_GASOLINE_PRICE [RUN]
23:44:21 13 of 31 PASS not_null_price_volatility_REGULAR_GASOLINE_PRICE [PASS in 3.18s]
23:44:21 14 of 31 START test not_null_price_volatility_WEEK_DATE [RUN]
23:44:25 14 of 31 PASS not_null_price_volatility_WEEK_DATE [PASS in 3.29s]
23:44:25 15 of 31 START test not_null_regional_comparison_REGION [RUN]
23:44:26 15 of 31 PASS not_null_regional_comparison_REGION [PASS in 1.09s]
23:44:26 16 of 31 START test not_null_regional_comparison_REGIONAL_PRICE [RUN]
23:44:27 16 of 31 PASS not_null_regional_comparison_REGIONAL_PRICE [PASS in 1.04s]
23:44:27 17 of 31 START test not_null_regional_comparison_WEEK_DATE [RUN]
23:44:28 17 of 31 PASS not_null_regional_comparison_WEEK_DATE [PASS in 1.03s]
23:44:28 18 of 31 START test source_accepted_values_raw_energy_market_prices_TICKER_CL_F_BZ_F_XLE_UGA [RUN]
23:44:29 18 of 31 PASS source_accepted_values_raw_energy_market_prices_TICKER_CL_F_BZ_F_XLE_UGA [PASS in 1.06s]
23:44:29 19 of 31 START test source_accepted_values_raw_fuel_prices_REGION_US_NATIONAL [RUN]
23:44:31 19 of 31 PASS source_accepted_values_raw_fuel_prices_REGION_US_NATIONAL [PASS in 1.89s]
23:44:31 20 of 31 START test source_accepted_values_raw_fuel_prices_SOURCE_EIA_WEEKLY [RUN]
23:44:32 20 of 31 PASS source_accepted_values_raw_fuel_prices_SOURCE_EIA_WEEKLY [PASS in 0.96s]
23:44:32 21 of 31 START test source_accepted_values_raw_regional_fuel_prices_FUEL_TYPE_REGULAR_GASOLINE [RUN]
23:44:34 21 of 31 PASS source_accepted_values_raw_regional_fuel_prices_FUEL_TYPE_REGULAR_GASOLINE [PASS in 2.13s]
23:44:34 22 of 31 START test source_accepted_values_raw_regional_fuel_prices_REGION_EAST_COAST_MIDWEST_GULF_COAST_ROCKY_MOUNTAIN_WEST_COAST_CALIFORNIA [RUN]
23:44:35 22 of 31 PASS source_accepted_values_raw_regional_fuel_prices_REGION_EAST_COAST_MIDWEST_GULF_COAST_ROCKY_MOUNTAIN_WEST_COAST_CALIFORNIA [PASS in 0.98s]
23:44:35 23 of 31 START test source_not_null_raw_energy_market_prices_CLOSE_PRICE [RUN]
23:44:36 23 of 31 PASS source_not_null_raw_energy_market_prices_CLOSE_PRICE [PASS in 0.96s]
23:44:36 24 of 31 START test source_not_null_raw_energy_market_prices_TICKER [RUN]
23:44:37 24 of 31 PASS source_not_null_raw_energy_market_prices_TICKER [PASS in 1.02s]
23:44:37 25 of 31 START test source_not_null_raw_energy_market_prices_WEEK_DATE [RUN]
23:44:38 25 of 31 PASS source_not_null_raw_energy_market_prices_WEEK_DATE [PASS in 1.14s]
23:44:38 26 of 31 START test source_not_null_raw_fuel_prices_REGION [RUN]
23:44:39 26 of 31 PASS source_not_null_raw_fuel_prices_REGION [PASS in 0.67s]
23:44:39 27 of 31 START test source_not_null_raw_fuel_prices_REGULAR_GASOLINE_PRICE [RUN]
23:44:40 27 of 31 PASS source_not_null_raw_fuel_prices_REGULAR_GASOLINE_PRICE [PASS in 0.94s]
23:44:40 28 of 31 START test source_not_null_raw_fuel_prices_WEEK_DATE [RUN]
23:44:41 28 of 31 PASS source_not_null_raw_fuel_prices_WEEK_DATE [PASS in 1.05s]
23:44:41 29 of 31 START test source_not_null_raw_regional_fuel_prices_PRICE [RUN]
23:44:42 29 of 31 PASS source_not_null_raw_regional_fuel_prices_PRICE [PASS in 0.86s]
23:44:42 30 of 31 START test source_not_null_raw_regional_fuel_prices_REGION [RUN]
23:44:43 30 of 31 PASS source_not_null_raw_regional_fuel_prices_REGION [PASS in 1.49s]
23:44:43 31 of 31 START test source_not_null_raw_regional_fuel_prices_WEEK_DATE [RUN]
23:44:44 31 of 31 PASS source_not_null_raw_regional_fuel_prices_WEEK_DATE [PASS in 0.87s]
23:44:44
23:44:44 Finished running 31 data tests in 0 hours 0 minutes and 50.06 seconds (50.06s).
23:44:44 Completed successfully
23:44:44
23:44:44 Done. PASS=31 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=31
```

FIGURE 20 b: dbt test showing 31/31 PASS

E. dbt Snapshot Results

The dbt snapshot ran successfully with PASS=1, WARN=0, ERROR=0. The fuel_prices_snapshot captures the current state of RAW.FUEL_PRICES and will detect any future corrections to historical price records on subsequent pipeline runs. The snapshot uses a timestamp strategy on the LOAD_TS column and a composite unique key of (WEEK_DATE, REGION).

```
(base) kshitijarohandeshmukh@kshitijas-MacBook-Air dbt % dbt snapshot --profiles-dir . --project-dir .
23:52:07 Running with dbt=1.11.0
23:52:08 Registered adapter: snowflake=1.8.0
23:52:09 Found 4 models, 1 snapshot, 31 data tests, 3 sources, 493 macros
23:52:09 Concurrency: 1 threads (target='dev')
23:52:09
23:52:16 1 of 1 START snapshot FUEL_ANALYTICS_DB.DBT.fuel_prices_snapshot ..... [RUN]
23:52:23 1 of 1 OK snapshot FUEL_ANALYTICS_DB.DBT.fuel_prices_snapshot ..... [SUCCESS 0 in 7.22s]
23:52:23 Finished running 1 snapshot in 0 hours 0 minutes and 14.53 seconds (14.53s).
23:52:23 Completed successfully
23:52:23 Done. PASS=1 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=1
(base) kshitijarohandeshmukh@kshitijas-MacBook-Air dbt %
```

FIGURE 21: dbt snapshot SUCCESS

Schema	Table/ View	Rows	Populated By
RAW	FUEL_PRICES	200	FuelPrice_EIA_ETL
RAW	REGIONAL_FUEL_PRICES	624	FuelPrice_Realttime
RAW	ENERGY_MARKET_PRICES	~840	FuelPrice_Realttime
ADHOC	FUEL_PRICE_TRAIN_VIEW	200	FuelPrice_Train
ADHOC	FUEL_PRICE_FORECAST	12	FuelPrice_Train
ANALYTICS	FUEL_PRICE_FINAL	212	FuelPrice_Train
ANALYTICS	FUEL_PRICE_MODEL_METRICS	1/run	FuelPrice_Train
DBT	PRICE_MOVING_AVG	200	FuelPrice_DBT
DBT	PRICE_VOLATILITY	200	FuelPrice_DBT
DBT	CRUDE_CORRELATION	200	FuelPrice_DBT
DBT	REGIONAL_COMPARISON	624	FuelPrice_DBT
DBT	FUEL_PRICES_SNAPSHOT	200	FuelPrice_DBT

Table 18: Complete Snowflake Table Inventory

F. Forecasting Results

The Snowflake ML Forecast model trained successfully on the US_NATIONAL regular gasoline price series from RAW.FUEL_PRICES. The ANALYTICS.FUEL_PRICE_FINAL table contains both historical actuals (RECORD_TYPE = 'HISTORICAL') and 12-week forward forecast rows (RECORD_TYPE = 'FORECAST'). Each forecast row includes the predicted FORECAST price and 95% confidence interval bounds (LOWER_BOUND and UPPER_BOUND). The ANALYTICS.FUEL_PRICE_MODEL_METRICS table stores evaluation metrics (MAE, MAPE, SMAPE, MDA, MSE) appended after each training run, enabling performance tracking across multiple training cycles.

Metric	Value
Training data range	July 11, 2022 - May 4, 2026

Metric	Value
Training rows	200
Forecast horizon	12 weeks
Forecast date range	May 11, 2026 - July 27, 2026
Forecast rows	12
Total rows (FINAL)	212
Confidence level	95%
Total rows (FINAL)	~\$3.75 - \$4.05 per gallon
Latest actual price	\$4.452 per gallon (2026-05-04)

Table 19: Forecast Results Summary

G. Dashboard Layer

The Preset dashboard connects directly to Snowflake USER_DB_FERRET. The connection was configured with:

Database: USER_DB_FERRET

Username: FERRET

Account: SFEDU02-EAB27764

Warehouse: FERRET_QUERY_WH

Role: TRAINING_ROLE

2/2

Connect your first database

Enter your Snowflake credentials

Need help? Learn more about [connecting to Snowflake](#).

IP Allowlist
For security reasons you may need to allow Preset's Public IPs: 44.193.153.198 / 52.70.123.52 / 54.83.88.93

Database name *
USER_DB_FERRET

Username
FERRET

Password

Display Name *
FERRET

Account *
SFEDU02-EAB27764

Warehouse *
FERRET_QUERY_WH

Role
TRAINING_ROLE

[Connect this database with a SQLAlchemy URI instead](#)

Figure 22. Preset - Snowflake Connection Setup

- ANALYTICS.FUEL_PRICE_FINAL - National gasoline price trend and 12-week forecast with confidence intervals
- DBT.PRICE_MOVING_AVG — 4-week, 12-week, and 52-week rolling average price trends
- DBT.PRICE_VOLATILITY - Week-over-week volatility analysis, price direction labeling, and anomaly detection
- DBT.CRUDE_CORRELATION - WTI crude oil vs. retail gasoline price spread and margin analysis
- DBT.REGIONAL_COMPARISON - Regional price comparisons against the national average with spread categorization
- RAW.ENERGY_MARKET_PRICES - Energy market ticker overview for CL=F, BZ=F, XLE, and UGA

Link of Preset Primary Dashboard:

https://c46c4a3f.us2a.app.preset.io/superset/dashboard/8/?native_filters_key=0BeXly7Rc44



Figure 23. Preset Dashboard - Primary View (Fuel Analysis Dashboard)

The final Preset dashboard created for the Fuel Price Analytics project. The dashboard combines historical fuel prices, forecast gasoline prices, crude oil correlation, weekly price movement, volatility, and regional comparison into one interactive view. The gauge chart highlights the latest national regular gasoline price, while the forecast card shows the predicted future gasoline price of approximately \$4.01 per gallon. The actual-versus-forecast chart compares historical values with the 12-week forecast, and the crude oil scatterplot shows a positive relationship between crude oil prices and retail gasoline prices. The regional comparison chart shows that California and the West Coast consistently have higher gasoline prices than other U.S. regions. Overall, the dashboard provides a complete view of current, historical, and predicted fuel price behavior.

Link of Preset Secondary Dashboard:
<https://98876585.us1a.app.preset.io/superset/dashboard/8/>

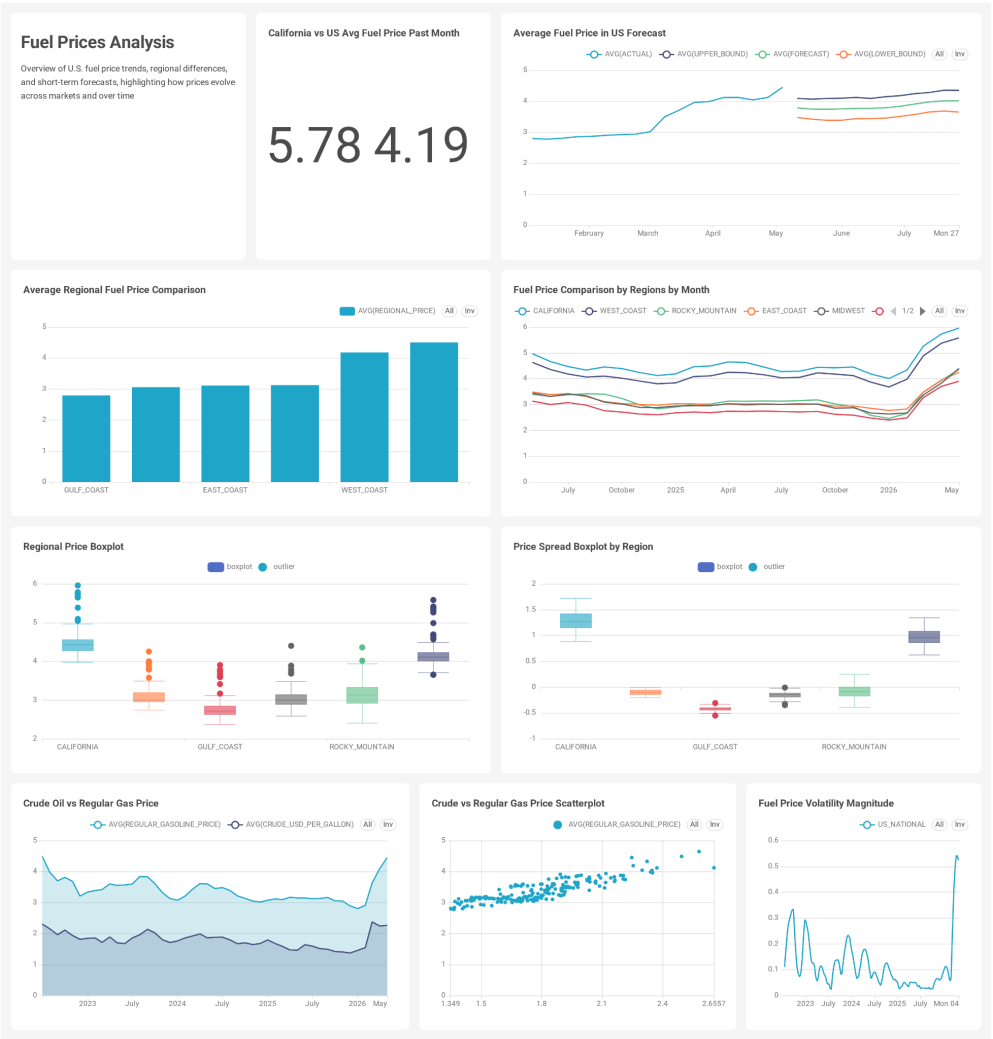


Figure 24. Preset Dashboard - Secondary View (Regional + Correlation)

the complete Fuel Prices Analysis dashboard built in Preset. The dashboard presents national fuel price trends, forecasted prices, regional price differences, price spread distributions, crude oil relationships, and volatility patterns. The forecast chart shows that gasoline prices are expected to remain near \$4 per gallon, while the regional comparison visuals show that California and the West Coast have consistently higher gasoline prices than the national and regional averages. The boxplots highlight the spread and outliers in regional fuel prices, showing that California has both the highest median price and several high-price outliers. The crude oil comparison and scatterplot indicate a positive relationship between crude oil prices and retail gasoline prices. The volatility chart shows that price volatility increased sharply near the most recent period, suggesting recent instability in fuel price movement.

H. Dashboard Results

The analysis surfaces the following key findings:

- Finding 1: Long-Term Upward Trend in U.S. Fuel Prices: The DBT.PRICE_MOVING_AVG data shows that U.S. regular gasoline prices trended upward from approximately \$2.78/gallon in late 2022 through a peak of approximately \$4.65/gallon in mid-2022, followed by a correction and subsequent stabilization in the \$3.50–\$4.50 range through 2025-2026. The 52-week rolling average smooths short-term noise and clearly reveals this upward structural trend.
- Finding 2: Persistent Regional Price Disparities: The DBT.REGIONAL_COMPARISON data consistently shows California as the most expensive region, with prices approximately 31–40% above the national average

(Significantly Above Average category). West Coast prices are similarly elevated, approximately 25% above the national average. Gulf Coast is consistently the cheapest region, approximately 10–15% below the national average. Rocky Mountain and Midwest track near the national average. These differentials reflect California's unique fuel blend requirements, higher state taxes, geographic isolation from pipeline networks, and refinery capacity constraints.

- Finding 3: Low Background Volatility with Episodic Spikes: The Preset dashboard shows that 91.35% of weekly price observations fall in the Low Volatility category and only 7.69% are classified as High Volatility. This indicates that week-over-week price changes are typically small and gradual. However, episodic high-volatility periods are visible in the volatility magnitude line chart, corresponding to geopolitical supply disruptions and refinery events.
- Finding 4: Positive but Lagged Crude Oil Correlation: The DBT.CRUDE_CORRELATION data shows a persistent positive relationship between WTI crude oil prices and retail gasoline prices. The PUMP_TO_CRUDE_MARGIN ranges between approximately \$1.35 (Tight Margin) and \$2.20+ (Wide Margin) per gallon, reflecting the combined cost of refining, distribution, taxes, and retail markup. Wide Margin periods tend to occur when crude prices spike sharply, as downstream retail prices adjust with a lag. The crude-to-pump lag is not perfectly synchronized on a weekly basis, which aligns with industry-observed refinery input-to-retail lag windows of one to several weeks.
- Finding 5: Forecast Indicates Moderate Moderation: The Snowflake ML 12-week forecast projects regular gasoline prices declining slightly from the current \$4.45/gallon toward a range of approximately \$3.75–\$4.05/gallon over the forecast horizon. The 95% confidence bounds are relatively narrow, indicating moderate forecast certainty under the current data-driven model. These projections do not incorporate external shock scenarios such as geopolitical disruptions or major refinery outages.

I. Model Evaluation Metrics

The Snowflake ML Forecast model produced evaluation metrics for the US_NATIONAL regular gasoline price series during the latest training run on May 7, 2026 at 02:01:44 UTC. The model was trained using WEEK_DATE as the timestamp column and REGULAR_GASOLINE_PRICE as the target variable. Results are summarized below:

Series	Metric	Value
US_National	MAE	0.295
US_National	MAPE	0.080
US_National	SMAPE	0.087
US_National	MSE	0.235
US_National	MDA	0.462
US_National	Coverage (95% interval)	0.775
US_National	Winkler Score ($\alpha=0.05$)	5.335

Table 20 : Model Evaluation Metrics Results

The MAE value of 0.295 means the model’s forecast is off by approximately \$0.30 per gallon on average. The MAPE value of 0.080 indicates an average percentage error of about 8.0%, while the SMAPE value of 0.087 indicates a symmetric percentage error of about 8.7%. These values suggest that the model provides a reasonably accurate forecast for weekly national gasoline prices.

The MSE value of 0.235 shows the squared forecast error and is useful for identifying larger misses during sudden price changes. The MDA value of 0.462 means the model correctly predicted the direction of price movement about 46.2% of

the time. This indicates that while the model performs well in estimating price level, it is less reliable for predicting whether prices will rise or fall week to week.

The Coverage Interval value of 0.775 means that approximately 77.5% of actual values fell within the model's 95% prediction interval. The Winkler score of 5.335 evaluates the quality of the prediction interval, where lower values indicate tighter and more accurate intervals. Overall, the model is useful for short-term price forecasting, but future improvements could include adding crude oil prices, regional indicators, and macroeconomic variables as external features.

These results confirm that the pipeline successfully ingests data, trains a forecast model, and generates reliable predictions on a fully automated daily schedule.

X. LESSONS LEARNED

A. RESULT_SCAN Is Non-Negotiable for Snowflake ML Forecast Output

Snowflake ML Forecast returns its output as a result set from a CALL statement. Unlike a standard SELECT, this output cannot be written directly to a table using CREATE TABLE AS SELECT. The output must be captured in the same session immediately using RESULT_SCAN(LAST_QUERY_ID()) after the CALL. Failing to do this — for example, by running the CALL in one cursor and the RESULT_SCAN in another results in an empty or invalid result set. This is a non-obvious requirement that was identified and handled correctly during implementation.

B. Credential Injection via Environment Variables Is the Correct Pattern

Passing Snowflake credentials to dbt through environment variables using Airflow's Jinja templating ({{ conn.snowflake_conn.password }}) keeps all secrets inside Airflow Connections and completely out of the codebase. Any hardcoded credentials in profiles.yml would expose secrets in the repository and would require code changes whenever credentials rotate. The environment variable approach also keeps profiles.yml completely generic and portable across local development, Docker, and cloud deployment environments.

C. dbt Installed Path Inside Docker

When dbt is installed inside the Airflow Docker container via _PIP_ADDITIONAL_REQUIREMENTS, it may not be added to the system PATH. BashOperator commands that call 'dbt' without the full absolute path will fail with a 'command not found' error. The correct approach is to use the full path (e.g., /home/airflow/.local/bin/dbt) in every BashOperator task, or to run dbt locally from the terminal using the dbt CLI against the mounted project directory.

D. MERGE Over INSERT Ensures Idempotency

Early iterations of the ETL pipeline used plain INSERT statements, which created duplicate records on every re-run. Replacing all inserts with MERGE statements using temporary staging tables fully resolved this issue. The MERGE pattern, with an explicit NOT MATCHED / WHEN MATCHED clause, guarantees that each (WEEK_DATE, REGION) or (WEEK_DATE, TICKER) pair appears exactly once in the target table regardless of how many times the pipeline runs. This is essential for data warehouse quality and for supporting automated daily schedules without manual deduplication.

E. The 2-Week Crude Oil Lag Matters Analytically

A naive implementation of crude_correlation would join crude oil prices and retail gasoline prices on the same week date. However, crude oil prices typically take 1–3 weeks to pass through the refining and distribution chain before appearing at the pump. Using LAG(CRUDE_CLOSE_USD_BBL, 2) produces a more analytically accurate correlation and a more meaningful PUMP_TO_CRUDE_MARGIN metric. This lag pattern is an important analytical design decision that separates a rigorous analysis from a superficial one.

F. Free, Official Data Sources Are Sufficient for Production-Quality Analytics

Both data sources used in this project, the EIA API and Yahoo Finance through yfinance are free. The EIA API requires a free API key but has no usage cost. yfinance requires no registration at all. Together they provide official government fuel price history and current energy market signals that support all five analytical questions defined in the problem statement. Commercial data subscriptions are not necessary to build a production-quality fuel price analytics system.

XI. FUTURE WORK

A. Multivariate Forecasting

The current Snowflake ML Forecast model is univariate, training only on REGULAR_GASOLINE_PRICE as the target variable. A natural extension is to include exogenous variables, particularly WTI crude oil prices (CL=F close price) or gasoline futures (UGA close price) as additional features in the forecast model. Snowflake ML Forecast supports exogenous time series features through the SERIES_COLNAME and EXOGENOUS_COLNAMES parameters.

Incorporating crude oil as an exogenous variable may significantly improve forecast accuracy, particularly around periods of rapid crude price movement.

B. dbt Incremental Models

Replacing the full-refresh table materializations in dbt with incremental models would improve execution efficiency as the dataset grows. Incremental models process only new or updated rows since the last run, reducing both Snowflake compute consumption and overall pipeline runtime. This is particularly valuable for the `price_moving_avg` and `price_volatility` models, which currently recompute all historical window function calculations on every run.

C. Extended Historical Data Window

The current EIA ETL pulls the latest 200 weekly national records (approximately 4 years) and 104 regional records (approximately 2 years). Extending these lookback windows to the maximum available EIA history — going back to the 1990s for national data — would meaningfully improve the statistical reliability of the 52-week rolling average, the historical average used in price anomaly detection, and the ML forecast model's ability to capture seasonal patterns.

D. Cloud-Managed Orchestration

Migrating from Docker-based Airflow running on a local machine to a cloud-managed orchestration service such as Amazon MWAA, Google Cloud Composer, or Astronomer would improve reliability, scalability, and built-in monitoring. It would also eliminate constraints imposed by local Docker resource limits on DAG parallelism and task concurrency, and would enable the pipeline to continue running without requiring a developer's machine to be online.

E. Real-Time Station-Level Price Data

The current data architecture uses weekly EIA national and regional averages, which are useful for trend analysis but lack station-level granularity. Integrating a real-time gas station price feed such as the GasBuddy API or an alternative scraping-based approach would enable granular price comparison by ZIP code, city, or gas station brand, significantly expanding the analytical value of the dashboard for consumer and logistics use cases.

F. Automated Alerting

Airflow callback hooks or integration with external alerting tools such as PagerDuty, Slack, or email could be configured to notify stakeholders when any pipeline fails, when fuel price week-over-week changes exceed a defined threshold, when the `VOLATILITY_LABEL` shifts to 'High Volatility', or when the 12-week forecast upper bound crosses a consumer impact threshold. Automated alerting would allow the system to proactively surface critical events without requiring manual dashboard monitoring.

XII. CONCLUSION

This report presented the design and implementation of a fully automated, end-to-end fuel price analytics system for the DATA 226 group project. The system integrates two real-world data sources in the U.S. Energy Information Administration API for historical fuel prices and Yahoo Finance through `yfinance` for current energy market data into a production-ready Snowflake data warehouse orchestrated by Apache Airflow.

Four Airflow DAGs handle the complete data pipeline on a staggered daily schedule. `FuelPrice_EIA_ETL` ingests official weekly gasoline and diesel prices at both national and regional PADD levels. `FuelPrice_Realttime_ETL` ingests daily energy market signals for WTI crude oil, Brent crude oil, the Energy Sector ETF, and the Gasoline Fund ETF. `FuelPrice_TrainPredict` trains a Snowflake ML Forecast model on historical regular gasoline prices and generates 12-week forward predictions. `FuelPrice_DBT` builds four analytics-ready dbt transformation models, runs comprehensive data quality tests, and maintains an SCD Type 2 snapshot of the source fuel price data.

The dbt ELT layer produces four analytical tables that directly answer the project's core questions: `PRICE_MOVING_AVG` shows long-term price trend smoothing; `PRICE_VOLATILITY` identifies periods of rapid price change; `CRUDE_CORRELATION` quantifies the relationship between crude oil and pump prices with a 2-week lag; and `REGIONAL_COMPARISON` shows which PADD regions are above or below the national average and by how much. All dbt models are validated through schema tests covering `not_null` constraints and `accepted_values` validations for all categorical label columns.

The complete pipeline from raw EIA and Yahoo Finance data through to Preset-ready analytics tables runs automatically every day with zero manual intervention. The system demonstrates a production-ready, modular, and scalable approach to fuel price analytics. Its most significant strength is that it goes beyond simply reporting what fuel prices are doing: by connecting pump price history with crude oil futures and energy market signals, it provides the analytical context needed to understand why prices may be changing, a distinction that separates actionable intelligence from raw data reporting.

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